

Review

Clean and Smart Energy Technologies for Agricultural Energy Internet Systems: A Comprehensive Review and Future Perspectives

Yuxin Wu and Xueqian Fu * 

College of Information and Electrical Engineering, China Agricultural University, Beijing 100083, China;
2023308130612@cau.edu.cn

* Correspondence: fuxueqian@cau.edu.cn

Abstract

The Agricultural Energy Internet (AEI) represents an emerging systemic paradigm driven by the convergence of intelligent agriculture and rural energy transformation. It is not a simple extension of agricultural informatization or rural electrification; rather, it redefines agricultural processes—such as irrigation, greenhouse environmental control, supplementary lighting, cold-chain logistics, and agricultural machinery—as perceptible, computable, and schedulable energy-related processes, thereby enabling the deep integration of agriculture, energy, environmental management, and intelligent decision-making. This review systematically examines the evolutionary trajectory of AEI, from early agricultural digitalization and Internet of Things (IoT)-based monitoring to edge intelligence and digital twin technologies, and ultimately to the coordinated optimization of agriculture–energy–environment systems. A comprehensive technical framework is established, encompassing physical energy coupling, multi-source sensing and actuation, interconnection and interoperability, edge–cloud collaborative control, data governance, digital twin modeling, artificial intelligence-enabled optimization, and application-oriented decision-making. The review further highlights that high-quality data governance, edge–cloud collaboration, and digital twin calibration are critical enablers of the transition from visualization-oriented management to closed-loop intelligent operation. In addition, this study clarifies the complementary relationship between agricultural informatization and electrification: the former provides capabilities for perception, prediction, optimization, and coordination, whereas the latter provides a controllable execution chain. Together, they constitute the foundation of a cyber-physical agricultural energy system. Finally, frontier research directions are identified, including high-temperature solid oxide electrolysis for hydrogen production, edge AI–IoT-enabled closed-loop agricultural operation, and privacy, security, and trust mechanisms in federated edge intelligence. The findings suggest that AEI can serve as a strategic technological framework for supporting the next generation of smart agriculture toward low-carbon, resilient, and collaborative operation.



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1. Background and Policy Context

1.1. Global Context

Over the past few years, intelligent agriculture has served as a critical approach to tackling issues surrounding food security, the sustainable growth of agriculture, and climate

change. According to data from the United Nations Food and Agriculture Organization (FAO), the global smart agriculture market is expected to witness a compound annual growth rate (CAGR) of over 15% during the 2021–2024 period, and in major agricultural nations, the adoption rates of technologies like the IoT, AI, and precision irrigation have risen from 12% to 27% [1]. The core objective of smart agriculture is to optimize resource allocation, reduce environmental impacts, and ensure food supply under global population growth through digital technologies. The Organisation for Economic Co-operation and Development (OECD) emphasized in a 2024 report that intelligent agricultural technologies can improve grain production efficiency by 20–30% while reducing fertilizer and water consumption by 15–25%, making them key drivers of agricultural transformation [2].

Developed countries in Europe and America have rolled out a series of smart agriculture policies to accelerate technological innovation and large-scale implementation. The European Commission updated its digital agriculture strategy in 2023 and set a goal to connect 80% of farms in EU Member States to the digital agriculture service platform by 2025. The strategy gives priority to precision agriculture, intelligent animal husbandry, and a data sharing system, and provides 7.2 billion euros of special funds for R&D and farmer training [3]. Back in 2024, the United States Department of Agriculture (USDA) rolled out a dedicated program targeting technological innovation and precision agriculture applications across the agricultural sector. They invested USD 5 billion to deploy Internet of Things devices on farms, develop artificial intelligence models, and realize cross-state data sharing. Their goal is to achieve 90% coverage of precision agriculture in major corn and soybean production areas by 2026 [4]. In 2022, Germany's Federal Ministry of Food and Agriculture introduced the Smart Agriculture 2030 initiative, focusing on sensor networks, autonomous agricultural machinery [5], and carbon footprint monitoring technologies, while encouraging SMEs to participate through tax incentives and subsidies.

Clean and efficient use of agricultural energy is the cornerstone of agricultural policies in the United States and Europe. The European Union unveiled its "Rural Renewable Energy Action Plan" in 2024, integrating agricultural energy transition into the framework of the "European Green Deal" and setting a target for renewable energy to account for 50% of agricultural energy consumption by 2030. The plan prioritizes supporting farm photovoltaic (PV) power stations, biomass energy utilization, and smart microgrids, while offering 30% investment subsidies for rural distributed PV projects [6]. The U.S. Inflation Reduction Act of 2023 allocated \$3.5 billion specifically for agricultural energy transition, including subsidies for purchasing clean energy equipment on farms, tax credits for renewable energy generation, and rural grid upgrades, encouraging farmers to adopt PV energy storage systems for energy self-sufficiency [7]. In 2023, the UK Department of Energy launched the "Rural Clean Energy Plan", investing £2 billion to support wind and PV projects in agricultural parks and promote flexible interconnection between agricultural energy systems and the main grid [8], with the aim to achieve 80% clean energy coverage in rural areas by 2025.

Currently, the advancement of intelligent agriculture in China has moved into the stage of extensive promotion and nationwide deployment, becoming a core pillar of rural revitalization and agricultural modernization. Data from the Ministry of Agriculture and Rural Affairs (MARA) shows that the total investment in China's smart agriculture sector topped 120 billion yuan over the 2022–2024 period. Emerging technologies like the Internet of Things, big data, and AI have been fully incorporated into the entire lifecycle of staple crops, including wheat, rice, and maize, spanning sowing [9], cultivation, field management, and crop harvesting.

To date, China has established 100 national smart agriculture demonstration bases, with precision sowing, smart irrigation, and green pest control technologies covering over

40% of major production areas. As rural digitalization continues to deepen, the integration of agricultural data resources is accelerating, and infrastructure such as the National Agricultural Product Quality and Safety Traceability Platform and the Agricultural and Rural Big Data Center is being progressively improved [9], providing robust support for the large-scale development of smart agriculture.

Recently, China has introduced a range of major policies aimed at establishing an institutional guarantee system for smart agriculture's development. In 2023, six departments including the Ministry of Agriculture and Rural Affairs and the Ministry of Industry and Information Technology (MIIT) jointly promulgated the "Smart Agriculture Development Action Plan (2023–2025)", which explicitly proposed to build 500 smart agriculture demonstration counties by 2025, cultivate more than 100 leading enterprises in smart agriculture, achieve a 30% digitalization rate across the entire production process of staple crops, and focus on supporting breakthroughs and industrialization of core technologies such as agricultural sensors [10], agricultural big data models, and Smart Agricultural machinery.

The 2024 Central Document No. 1 emphasized "accelerating smart agriculture development and advancing agricultural digital transformation", proposing the establishment of a national agricultural data resource system and the extensive application of technologies, including satellite remote sensing and unmanned aerial vehicle observation, to elevate agricultural operational efficiency [11]. Additionally, the 2023 Digital Rural Development Guidelines were released by the National Rural Revitalization Administration (NRRRA). This policy regards smart agriculture as a core element of digital rural development. It also encourages the construction of digital agricultural service stations in rural regions [12]. These measures aim to enhance farmers' digital application skills.

The transition of agricultural energy is an important starting point for China's rural energy revolution and green agricultural development, with relevant policies mainly focusing on clean energy substitution and efficient utilization. Released in 2024, the 2024–2027 Implementation Plan for Rural Energy Revitalization was officially promulgated by the National Development and Reform Commission (NDRC) together with the National Energy Administration (NEA) in a joint effort. The plan clearly sets a target for 2027. By that year [13], the share of renewable energy consumption in rural regions is projected to reach 25 percent.

Key priorities of this scheme include county-scale distributed PV expansion, integrated exploitation of rural biomass energy, and intelligent microgrid development in agricultural parks, alongside kWh subsidies for rural PV and energy storage constructions. The National Energy Administration jointly initiated the Rural Energy Comprehensive Demonstration Project with the Ministry of Agriculture and Rural Affairs in 2023. The project promoted technologies such as PV-equipped greenhouses, livestock manure biogas production, and agricultural machinery electrification in 200 counties across the country. Up to now, more than 300,000 rural distributed PV projects have been established [14]. Their total installed capacity has exceeded 150 million kilowatts.

In 2022, the NEA released the document "Guidance on Accelerating the Transformation and Development of Rural Energy". The document explicitly supports clean energy alternatives for agricultural production. It also encourages integrated models [14], such as agricultural PV and wind-power complementarity. In addition, it promotes the deep integration of rural energy systems with agricultural production.

Globally, smart agriculture has become a core direction of agricultural transformation. Countries around the world have promoted their development through policy guidance, capital investment, and technological innovation, while the clean and efficient utilization of agricultural energy has become an important supporting trend. Keeping pace with international developments, China has advanced smart agriculture in light of its own agri-

cultural conditions. By improving policy systems, building demonstration platforms, and strengthening key technological breakthroughs, China has achieved an organic integration of large-scale promotion and localized innovation. In the future, deeper technological exchange and sustained policy support will further integrate smart agriculture with agricultural energy transition, laying a stronger foundation for global agricultural sustainability and China's rural revitalization.

1.2. Review Methodology and Analytical Criteria

To improve the transparency of the research methodology, this review divides the literature review process into four consecutive stages: literature identification, preliminary screening, eligibility assessment, and analytical coding. The academic databases used for formal retrieval include the Web of Science Core Collection, Scopus, IEEE Xplore, ScienceDirect, and SpringerLink. Google Scholar is only adopted for forward and backward citation tracking, as well as the supplementary retrieval of early basic research. In addition, this paper cites authoritative policy documents and statistical reports released by the Food and Agriculture Organization, the Organisation for Economic Co-Operation and Development, the European Commission, the United States Department of Agriculture, the Ministry of Agriculture and Rural Affairs, the National Development and Reform Commission, and the National Energy Administration. These materials are only used as background references and not included in the core technical evidence. Within the scope of database functions, the retrieval covers titles, abstracts, and keywords. A total of more than 170 literature records were obtained through initial retrieval and citation tracking. After deduplication, title and abstract screening, and full-text eligibility assessment, 143 studies were finally retained as the core technical literature, with policy documents supplemented as contextual background materials.

The final formal database retrieval and citation-tracking update were completed on 30 April 2026. For reproducibility, the core Boolean search strings were applied to titles, abstracts, and keywords where supported by each database:

(S1) ("Agricultural Energy Internet" OR "rural energy internet" OR "agricultural energy system" OR "farm energy system") AND ("smart agriculture" OR "precision agriculture" OR "agricultural IoT" OR "digital agriculture").

(S2) ("agricultural electrification" OR "electric agricultural machinery" OR "greenhouse load" OR irrigation OR "cold chain") AND (photovoltaic OR renewable OR microgrid OR "energy storage" OR "demand response").

(S3) ("agricultural digital twin" OR "edge computing" OR "edge intelligence" OR "AIoT" OR "closed-loop control") AND (greenhouse OR irrigation OR livestock OR farm).

(S4) ("agricultural microgrid" OR "rural microgrid" OR "carbon-constrained dispatch" OR "virtual power plant") AND (agriculture OR greenhouse OR irrigation OR rural). Equivalent syntax adjustments were made only when a database required different wildcard or quotation rules, while the logical meaning of each query was kept unchanged.

The extraction of key research findings in Section 2 adopts a standardized coding procedure. After removing duplicate literature, titles, and abstracts are screened according to three inclusion criteria: direct relevance to agricultural energy systems or smart agriculture, clear technical content rather than simple policy comments, and interpretable research contributions to system architecture, control strategies, energy utilization, or data-driven decision-making. Subsequently, literature is sorted and coded from multiple dimensions, including application scenarios, energy/resource types, agricultural production links, research target issues, research methods, evaluation indicators, main research findings, and research limitations. The comparative descriptions in Tables 1–3 and the comprehensive

discussion paragraphs in Section 2 are summarized based on the coding matrix, rather than subjective induction merely by reading the literature.

The core review time window of this paper is set to 2020–2025. This period witnesses the in-depth integration of agricultural electrification, agricultural Internet of Things, edge intelligence, digital twins, and low-carbon scheduling technologies, representing a technically symbolic stage for the development of the Agricultural Energy Internet (AEI) paradigm. Therefore, Sections 2–4 comment on recent technological progress based on this five-year timeframe. Early milestone studies are retained only when necessary to support the 25-year developmental context of the field, summarized in Table 4 and Section 3. This review does not replace the complete 25-year evolutionary process with literature published from 2020 to 2025. Instead, recent studies are used for horizontal comparison of the current status, while classic milestone studies serve as historical anchors for development.

This paper conducts retrieval by combining keyword search with backward citation tracking of representative reviews and classic studies at each stage. The retrieval focuses on stage-specific keywords, including precision agriculture, agricultural Internet of Things, edge/fog computing, digital twins, trusted collaboration, microgrid energy management, and carbon-constrained agricultural energy systems. Table 4 adopts a differentiated retrieval logic. Its research objective is not to compare state-of-the-art methods, but to restore the evolutionary process of the field at each stage. Candidate literatures are included in Table 4 only when they can represent the transformation of research paradigms, iteration of core technical systems, or upgrading of system functions.

To establish clear and unified comparison criteria, this paper adopts a general analytical perspective, covering the core problems to be solved, controllability of agricultural production processes, sources of uncertainty, optimization objectives, decision-making time scales, as well as evaluation indicators such as cost, carbon emissions, power supply reliability, power quality, and agricultural production benefits. This unified analytical framework is applied consistently in Sections 2, 3, 5 and 6 to realize structured comparison and comprehensive review.

To implement the above screening criteria, this review further classifies literature evidence into six quantifiable dimensions: (1) agricultural production types, (2) energy forms, (3) load-curve characteristics, (4) reliability and safety indicators, (5) environmental indicators, and (6) optimization objectives. This classification avoids purely descriptive comparisons and improves the objectivity and transparency of the review conclusions.

Meanwhile, to avoid citation mismatch, this paper implements a citation-consistency audit. Each in-text citation is cross-checked against the title and abstract of the corresponding reference. Special emphasis is placed on verifying statements related to agrivoltaic shading and crop-yield trade-offs, techno-economic sensitivity analysis of greenhouses, microgrid operation regulation, and greenhouse sensor networks. Only citations that directly support the corresponding statements are retained. All figures in this paper are independently drawn and compiled by the authors based on the literature review and the references cited in the corresponding sections; they are not directly reproduced from any single external source.

Table 1. Representative studies on agricultural electrification technology.

Ref	Year Range	Research Focus	Key Findings
[15,16]	2022–2024	Agricultural PV and agricultural PV complementary energy supply system	Agrioltaics improves land use and cultivation; integrates PV with microgrids for stable green power
[17–21]	2020–2025	Power prediction, scenario simulation, and grid connection planning of agricultural new energy	Enhanced agri-energy system: prediction, simulation, efficiency, planning
[22–26]	2020–2025	Rural electrification, microgrid, and distribution network operation guarantee	Improved reliability and efficiency of agricultural power supply
[27–29]	2023–2025	Electrification of smart agricultural machinery and control of operation equipment	Optimized agricultural machinery via MPC and GNSS/PTO
[30,31]	2022–2023	Smart farm energy monitoring and greenhouse autonomous control	PPC closed-loop enables power anomaly detection and adaptive greenhouse control
[32–35]	2012–2025	Low power perception and communication infrastructure of Agricultural Internet of Things	Enhanced sensing and communication for agricultural electrification

1.3. Distinct Contributions Relative to Recent Reviews

Existing relevant reviews have discussed the fundamental theories of AEI, controlled environment agriculture, agricultural digital twins, and closely coupled smart farming systems. Accordingly, this paper does not claim innovation by merely enumerating more technologies. Instead, its incremental contribution lies in clearly defining the system boundary of AEI and synthesizing existing literature from the perspective of the information-energy closed loop.

Specifically, this review defines AEI as a cyber-physical energy system. Within such a system, agricultural production processes are transformed into flexible loads constrained by agronomic rules; distributed energy resources evolve into schedulable production assets; artificial intelligence and digital twin modules realize prediction, verification, and optimal scheduling under multiple constraints, including cost, carbon emissions, power supply reliability, power quality, and agricultural output.

This research framework distinguishes this review from previous studies that merely focus separately on smart agricultural information systems, rural renewable energy deployment, or standalone digital twin modeling.

2. Literature, Landscape, and Gaps

2.1. Investigation on Agricultural Electrification Technology

2.1.1. Existing Deficiencies in Agricultural Electrification Technology

The development of agricultural electrification is relatively fast, but key obstacles hinder its large-scale application. Mainly, the uncertainty of the coupling between agricultural power demand and PV power generation makes energy management complicated. Agricultural power demand has seasonal characteristics and large fluctuations, and PV power generation depends on weather and has intermittency. The bilateral uncertainty presents enormous difficulties for the reliable operation of agricultural microgrid systems [23]. High energy storage cost is a major problem. It is necessary to use battery energy storage to resolve PV fluctuations and also improve the reliability of the system. However, its initial

investment is relatively large, which adds an economic burden to agricultural users with low profits [23]. In addition, there is a lack of a unified framework to balance power generation, crop production, and ecological impacts [15]. Also, the network security research of agricultural energy systems is not sufficient; after the popularization of smart inverters and networked control, there is a risk of network attacks, and a strong distributed resilient control strategy is needed [36].

To solve the problems of coupling instability and high physical energy storage cost, recent research has transformed static energy management into dynamic, scenario-based energy storage optimization. For example, Fu X et al. developed a knowledge-integrated generative adversarial network (GAN) to simulate the energy storage time series of the annual weather and promote PV integration analysis [37]. They reconstructed a new GAN architecture using bidirectional long short-term memory network (Bi-LSTM) and style conversion to achieve accurate PV time dynamics [38]. In order to efficiently manage a large amount of scenario data, a method of sharing weight graphs and perceptible variation clustering is adopted [19]. In addition, the adoption of statistical machine learning is encouraged to perform power flow analysis affected by meteorological conditions [17]. These solutions show that implementing high-fidelity prediction and scenario extraction algorithms is more economically feasible than blindly expanding battery storage.

The reliance of agricultural production on power supply reliability should be regarded as a production risk issue rather than a pure power supply problem. For instance, power outages in irrigation systems lead to crop water shortage; supply interruptions in cold chains easily cause agricultural product spoilage; and blackouts in greenhouses rapidly destabilize the internal microclimate. Therefore, the reliability assessment of AEI should incorporate indicators such as unavailable alternative energy conditions, maximum allowable downtime, expected economic loss of agricultural products, and the priority of emergency loads.

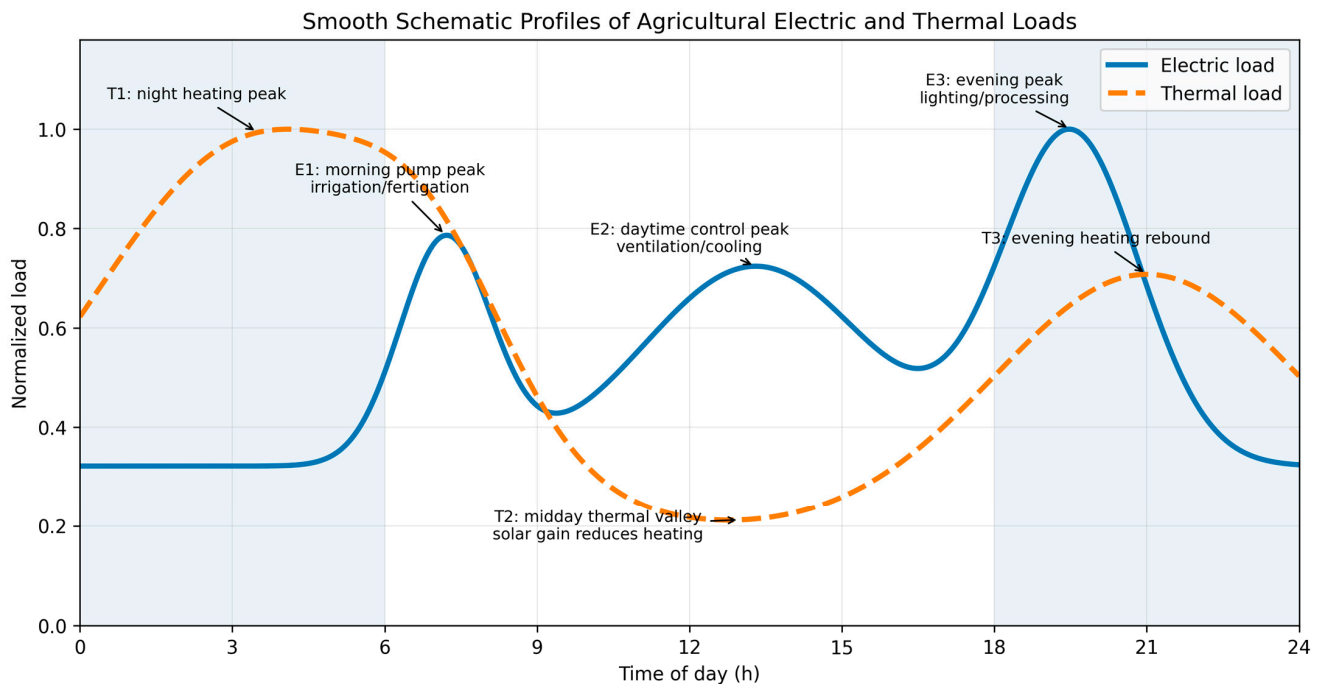
From the perspective of agricultural production processes, the load profile of AEI systems cannot be simply characterized by annual electricity consumption alone. Irrigation pumps usually generate short-duration high-power peak loads in dry seasons; greenhouses form coupled electric-thermal loads through heating, ventilation, cooling, supplementary lighting, and carbon dioxide regulation; cold-chain warehousing requires a continuous power supply with extremely low outage tolerance; drying and processing procedures produce short-term process loads during the harvest season.

Accordingly, future research on AEI should standardize the reporting of electric load curves, thermal load curves, and process energy intensity indicators. Typical metrics include electricity consumption per ton of crops, electricity consumption per square meter of greenhouse, electricity consumption per cubic meter of irrigation water, and heat demand per unit product. Figure 1 provides a schematic example of typical daily electric and thermal load profiles for representative agricultural enterprises, illustrating why AEI studies should analyze temporal load shapes rather than annual energy totals alone.

2.1.2. Development Trends of Agricultural Electrification Technology

In order to meet these challenges, agricultural electrification technology is developing toward intelligent, integrated, and accurate solutions. This is driven by several key trends. The most important trend is to deepen and diversify the agricultural system, go beyond the simple mode of “power generation at the top, planting at the bottom”, and adopt a more complex system design. This progress includes developing translucent PV materials with adjustable spectra to meet the light needs of different crops, optimizing PV support structures to better coordinate power generation and agricultural activities, and expanding agricultural applications to aquaculture, forestry, and animal husbandry [15].

The final aim is to construct a standardized appraisal and design paradigm for agricultural photovoltaic systems.



Note: Smooth normalized schematic profiles based on typical agricultural load mechanisms; not measured enterprise data.

Figure 1. Schematic profiles of agricultural electric and thermal loads.

The deep integration of artificial intelligence and data-driven technology has become the core trend of future development. It can resolve the coupled uncertain characteristics of energy provision and power consumption with a viable technical scheme. This convergence is mainly manifested in the following three aspects: prediction, optimization, and diagnosis. In the prediction field, advanced deep learning models such as bidirectional long and short-term memory time-domain convolution network (TCN bilstm) and spatio-temporal graph convolution network (st-gcn) have been applied to the ultra-short-term to short-term prediction of PV power generation [36] and electricity load prediction [24] providing precise data support for energy dispatching. In the optimization domain, algorithms like model predictive control (MPC) [29] and reinforcement learning (RL) are utilized to achieve real-time optimal energy management in agricultural microgrids and hybrid-powered agricultural machinery, maximizing economic benefits under multiple constraints. In the diagnosis domain, AI-based anomaly detection technology will be applied to electricity monitoring in smart farms [31], enabling early fault warning and precise operation and maintenance.

Comparing the reviewed methods by problem type can clearly delineate roles. Predictive models are most effective for renewable energy intermittency and load uncertainty, but they themselves cannot solve execution constraints. Model predictive control is more suitable for greenhouse, irrigation, and mechanical scenarios. In these scenarios, the state is measurable, and the drive is continuous. Random or robust optimization is more suitable for microgrid and park-level scheduling under coupled uncertainty of weather, price, and demand. The method based on digital twins is most powerful when strategy validation, risk drills, and cross-domain calibration are required. From this, it can be seen that the key issue is not which method is best globally, but which method fits the structure of the agricultural energy problem being solved.

Furthermore, rural transportation electrification is increasingly integrating with Demand Side Response (DSR). Agricultural machinery such as electric tractors and sprayers, along with rural electric vehicles, are no longer merely electrical devices in the traditional sense but have evolved into mobile distributed energy storage units capable of participating in Vehicle-to-Grid (V2G) interactions. Through properly formulated power pricing schemes and DSR strategies, these flexible load resources can be scheduled to recharge under high PV generation conditions and inject electricity into the grid during periods of peak electricity demand. This effectively smooths the net load curve and enhances rural power grids' capacity to accommodate renewable energy. More broadly, coordinated optimization of coupled power-communication networks provides a useful basis for dispatching large-scale flexible loads as operating reserve resources in AEI-oriented rural grids [39].

Another notable development trend is the rise of flexible interconnection and DC microgrids, which have become crucial solutions to address the shortcomings of rural power grids. Building flexible interconnected microgrids based on smart inverters and power electronic transformers (PETs) has emerged as a key development direction. DC microgrids, in particular, demonstrate significant potential in agricultural scenarios by enabling efficient and convenient integration of PV systems, energy storage, and DC loads (such as agricultural LED lighting). Research in this field focuses on developing safe, stable, and highly resilient distributed control strategies.

Notably, breakthroughs in ultra-low-power IoT and wireless energy technologies are pivotal for advancing precision agriculture, as ultra-low-power IoT infrastructure is essential for large-scale, long-term, and low-cost monitoring of farmland and greenhouse environments. Key trends in this field include: designing nanowatt-scale power-consumption sensor nodes and communication protocols [33], exploring radio frequency (RF) energy-harvesting wireless technologies, including metamaterial-network-based systems for integrated wireless power transfer, sensing, and communication [40], and developing drone-assisted data acquisition and energy supply technologies [41] to provide continuous power and data support for distributed sensors in farmland.

System security and resilience are now receiving growing attention in agricultural power systems, with cybersecurity protection emerging as a key development focus. Current research trends include: developing distributed mobile target defense (MTD) frameworks to counter false data injection attacks [26], and building resilient control systems capable of real-time detection, assessment, and compensation for cyberattacks [36], ensuring core agricultural power infrastructure can maintain basic operations even when under attack.

The fundamental limitation of agricultural electrification is the severe spatiotemporal mismatch between intermittent PV supply and rigid agricultural demand, which cannot be economically solved by battery storage alone [23]. However, while such algorithmic optimizations successfully resolve the temporal dispatch challenges of electrification, the actual physical deployment of these distributed energy assets inevitably triggers spatial and agronomic conflicts. In the future, it is necessary to resolve the spatial and agronomic conflicts caused by the deployment of distributed energy while implementing optimization algorithms, so as to promote the high-quality development of agricultural electrification.

2.2. Overview of Rural Energy Technologies

At the parcel-planning scale, optimal PV and agrivoltaic site selection should also incorporate physical and socioeconomic constraints, so rural energy deployment should be treated as a spatial planning problem rather than a simple capacity-expansion problem [42]. Table 2 summarizes the rural energy technology studies reviewed in this subsection, covering agrivoltaic planning, crop-response mechanisms, integrated energy systems, storage

and multi-energy coupling, irrigation infrastructure, compound land use, and agricultural equipment electrification.

Table 2. Representative studies on rural energy technologies.

Ref	Year Range	Research Emphasis	Key Findings
[43–47]	2022–2025	Agricultural photovoltaic system control, configuration optimization, and site selection planning	Tech-space-economy integrated agrivoltaics for improved performance and yield
[48–51]	2020–2025	Crop monitoring, microenvironment response, and yield mechanism of agroluminal system	Effects of PV shading and microenvironment on crops; Shading key of output power balance
[52–54]	2024–2025	Integrated energy system of facility agriculture and low-carbon operation	Upgraded agricultural energy: integrated, low-carbon, multi-energy
[55,56]	2023	PV thermal coupling and cascade utilization of solar energy	Optimized solar PV-T systems for agri-drying, heating, and irrigation
[57]	2025	Agricultural water energy synergy and PV irrigation infrastructure	Integrated agricultural energy for irrigation and sustainable power
[58–60]	2022–2025	Compound land use, ecosystem services, and farmland environment restoration	Multi-functional land use for energy, remediation, and governance
[61]	2021	Electrification of agricultural equipment and PV terminal replacement	PV/renewable energy to replace diesel and promote low-emission agriculture

2.2.1. Current Status and Development Trends of Rural Energy Technologies

(1) Economic Viability and Business Model Vulnerability

Many rural energy projects, particularly greenhouse systems, show significant parameter sensitivity in economic performance. Minor changes in retail power prices, grid feed-in tariffs, agricultural yields, and PV component expenses can result in dramatic shifts in Net Present Value (NPV) and investment recovery cycles. More importantly, when the yield of individual crops decreases to a critical threshold, the NPV of the project may become negative. This shows that farm management practices and yield stability are the key factors determining the feasibility of investment.

The fluctuation of PV module price also plays an important role. The same study shows that the price reduction can greatly shorten the payback period. When the price exceeds the critical threshold, the project may lose all economic feasibility. The profitability of the large-scale agricultural and fishery power joint production system is closely related to regional factors, such as solar energy resource allocation and on-grid electricity price policy [45]. This means replicating identical technical solutions across different regions carries considerable uncertainty.

(2) Planning, Siting, and Land-Use Challenges

The success of rural renewable energy projects depends on strategic site selection and supportive land-use policies. The PV project is a multi-criteria decision-making challenge. It is necessary to conduct a comprehensive assessment and risk assessment on climate conditions, environmental factors, geographical location, economic benefits, and social impacts. The traditional analytic hierarchy process (AHP) often faces the problems of conflict and redundancy when dealing with numerous criteria. To address this, scholars have proposed integrated approaches like Weighted Association and Product Assessment System (WASPAS), which combines correlation weighting with weighted aggregation and product evaluation, to better balance qualitative and quantitative factors.

At the institutional level, conflicts often arise between farmland protection and energy development. For instance, non-agricultural activities in agricultural protected areas are permitted only when conditions such as “absence of viable alternatives” and “no significant negative impacts” are met [46]. Furthermore, the legal framework governing distributed self-consumption, power sharing, and sales, along with capacity thresholds and approval procedures, directly determines the organizational structure and trading models of rural energy projects.

(3) Agrivoltaic Integration in Solar Greenhouses

A key technical bottleneck lies in the shading impact caused by photovoltaic panels, which changes the intensity and spatial uniformity of PAR (400–700 nm) and thus exerts a direct influence on crop development [48]. Severe shading not only reduces the availability but also leads to uneven radiation distribution within the planting space. Quantitative studies further demonstrate that increasing PV panel coverage significantly exacerbates radiation heterogeneity within greenhouses. Broader literature analysis indicates that radiation intensity beneath PV panels may decrease by over 40%, and coverage levels between 50 and 100% typically inhibit the growth and development of most crops, except for shade-tolerant crops or specific varieties [49]. When PV panel coverage increases from 50% to 100%, the growth and yield of most crops are markedly restricted; whereas lower coverage (e.g., 25%) is more likely to keep yield declines within acceptable limits [50]. These findings suggest that pursuing excessively high PV installation densities may compromise the primary agricultural functions of such systems.

(4) Resource Intermittency and Supply–Demand Imbalance

Renewable energy in rural areas, led by PV and wind power generation, is intermittent in nature and is extremely vulnerable to seasonal fluctuations. In contrast, agricultural energy demand (including irrigation, environmental regulation, cold chain logistics, and agricultural product processing) often shows rigid characteristics and is concentrated in a specific time period. This time mismatch will lead to the coexistence of renewable energy reduction and energy shortage. This challenge is particularly prominent in conservation agriculture (such as greenhouse cultivation), where power demand is huge and highly sensitive to meteorological conditions and ventilation dynamics. Without a dynamic thermal model, system scale design and energy matching may deviate significantly [53], ultimately leading to two issues: overconstruction and unstable energy supply. Although energy storage technologies can alleviate such fluctuations in agricultural applications to a certain extent, they usually require a large amount of initial investment and extend the payback period. For example, integrating a dual tank heat storage system into some rural integrated energy systems has been shown to significantly increase the total project cost and discounted payback period [54,56].

In addition to the photovoltaic-led approach, AEI should also clearly consider locally matched hybrid combinations, such as small-scale hydropower allowed by wind energy, terrain, and water resources, biomass/biogas from crop residues and livestock manure, and their coordinated operation with storage and main power grids. Therefore, the technological focus should shift from a narrative of renewable energy from a single source to multi-energy coupling in specific scenarios.

(5) Slow Electrification of Agricultural Machinery

Rural energy transformation requires not only decarbonization at the supply side, but also electrification of terminal equipment. More and more people are beginning to agree that new energy agricultural machinery, such as electric tractors, is the key to realizing low-carbon agriculture. However, empirical research shows that large-scale adoption will face

multiple constraints, including high initial cost, insufficient power performance, limited range, and operating costs related to battery life. Although farmers can accept a moderate premium, they are generally unwilling to bear high procurement costs or high monthly operating expenses [59]. This shows that policy tools (especially agricultural subsidies) are still crucial for promoting the promotion of innovative agricultural machinery.

This transformation is equally important for livestock systems, where ventilation, heating, cooling, milking, water supply, and manure treatment form highly energy-dependent processes. In this case, agricultural waste disposal is not a peripheral issue. Feces, straw, and agricultural processing residues can be converted into biogas, heat, electricity, digestive fertilizers, or biochar, thereby linking waste management, energy substitution, and carbon reduction in a comprehensive AEI pathway.

(6) Operation and Maintenance Constraints and Long-Term Sustainability

Technical feasibility in rural areas does not automatically translate into sustainable long-term business. The distributed energy system in rural areas often faces challenges such as the shortage of maintenance personnel, the extension of the equipment replacement cycle, and the instability of the supply chain. Taking the PV irrigation system as an example, the implementation of the project requires proper configuration and capacity matching among PV modules, inverters, and batteries, and factors such as temperature loss, line loss, and battery efficiency must be taken into account [62]. Without standardized operation and maintenance processes and remote monitoring, it will be difficult to ensure long-term stable operation and full life cycle cost control.

This problem is particularly prominent in conservation oriented agriculture, in which the coupling use of fans, heat pumps and other equipment increases the complexity of the whole system [54]. Without dynamic modeling and continuous calibration of key processes such as osmotic ventilation and radiant heat gain, it is difficult for the operation strategy to maintain the best performance for a long time.

In addition to operational efficiency, rural energy systems should also be evaluated within broader environmental boundaries. This includes not only on-site emissions and resource consumption, but also the manufacturing, replacement, recycling, and disposal of photovoltaic modules, batteries, inverters, power electronic equipment, and other storage/conversion devices. Incorporating environmental burden into the lifecycle helps avoid transferring carbon or pollution from farm boundaries to equipment supply chains.

The environmental boundaries of AEI should not only cover production-related factors such as fertilizers, water, and land use. It should also cover energy-related environmental burdens related to equipment manufacturing, replacement, and disposal. Photovoltaic modules, inverters, batteries, heat pumps, and power electronic converters may transfer emissions and pollution from the farm operation phase to the supply chain or end-of-life phase. Therefore, when evaluating the sustainability of AEI, life cycle assessment, recycling responsibility, secondary use, and safe disposal of batteries, and on-site carbon reduction should be considered.

Furthermore, the scope of AEI should not be limited to crop production alone. The livestock system involves highly energy-dependent processes such as ventilation, heating, cooling, milking, drinking water supply, manure treatment, and epidemic prevention monitoring. The reliability level and load characteristics of these processes are different from those of crop systems. It can be seen that animal husbandry production is the basic scenario for AEI application, rather than a peripheral extension.

A more complete AEI framework should include agricultural waste collection, pre-treatment, anaerobic digestion, biogas purification, cogeneration, digestate recovery, and biochar production. This approach transforms straw, manure, and processing residues from

environmental burdens into dispatchable energy and soil improvement resources, thereby linking waste management, rural energy substitution, and carbon footprint reduction.

The deployment of rural energy systems faces the bottleneck of conflicts between PV shading and crop yield [50]. The sensitive economic models and complex operation and maintenance problems in traditional agriculture make this situation worse [53]. For developing sustainable agricultural microgrids, resolving physical space limitations and shifting to active optical control and multi-energy synergy technologies is imperative.

2.2.2. Development Trends of Rural Energy Technology

(1) Integrated Rural Energy Systems

A key shift in the development of rural energy technology is from independent power generation equipment to embedded infrastructure in agricultural production systems. This transformation emphasizes the comprehensive utilization of land, water resources, and facility space, while pursuing cumulative economic benefits. In this context, the agricultural and aquaculture power system is expected to achieve higher standardization and scalability through demonstration projects, thus forming a replicable development model [45]. PV greenhouses have transformed from small schemes for simple power generation into small projects for customized light environments. By optimizing PV coverage, increasing greenhouse height, and integrating auxiliary lighting with crop layout plans, these systems are actively pursuing stable production and increased agricultural added value rather than passively accepting yield losses [50].

(2) PV Innovation for Conservation Agriculture

With the aim of fundamentally addressing the trade-off between PV shading and crop yield, subsequent studies will place greater emphasis on translucent and spectrally selective PV materials. They can actively adjust greenhouse lighting, rather than being passively limited. Existing studies have shown that the main limitations of crystalline silicon PV modules are their sensitivity to photosynthetic active radiation (PAR) attenuation and uneven distribution caused by shading. Therefore, emerging alternatives, such as thin-film PV, organic photovoltaic (OPV), spectral selective design, concentrating optical system, and luminous solar concentrator, have been identified as promising development directions, although there are challenges in terms of cost and conversion efficiency [49]. OPV can be absorbed and customized by material design to meet the needs of different optical transmissions. Concurrently, performance evaluation is advancing from rough estimation to a combination of simulation and physical measurement, utilizing improved incidence angle and temperature corrections to provide reliable quantitative support [49].

(3) Storage and Multi-Energy Synergy

Considering the high cost and short life of battery energy storage, the solution in rural areas is to combine energy storage technology with heat pump technology. Transform the main agricultural energy demand into scheduled loads to reduce the dependence on electrochemical energy storage. Although heat storage increases the initial investment and affects the return of the investment cycle of the project, its advantage is to extend the operation time of the equipment and smooth the fluctuation of solar energy supply [56]. Therefore, future optimization efforts will focus on thermal storage materials, working fluid selection, and system-level techno-economic design.

In parallel, the synergistic optimization of wind-solar hybrid systems with agricultural loads has emerged as a research hotspot. In high-load scenarios such as greenhouses, large-scale wind turbines, PV systems, and heat pumps can be integrated into a unified energy framework, and regional differences in wind-resource capacity factors strongly affect system configuration and dispatch optimization [53]. Therefore, future technolog-

ical development should prioritize location-specific technology portfolios and dispatch strategies rather than simply replicating standardized solutions across regions.

(4) Data-Driven Deployment and O&M

The Agricultural Internet of Things (Agri-IoT) underpins rural energy systems by turning isolated hardware into computable, optimized, and commercially viable infrastructure. Within this framework, intelligent site selection and planning methods must integrate climate conditions, topography, ecological environment, transportation accessibility, grid connection conditions, and social risks into a unified multi-criteria decision-making system [62].

In practical applications, the importance of online monitoring and predictive maintenance has become increasingly prominent, especially for PV irrigation systems and greenhouse energy systems. The platform-based operation and maintenance solution uses sensor technology to realize remote diagnosis, fault warning, and equipment aging identification, which can effectively reduce downtime and hidden costs caused by insufficient local technical capacity [62]. For PV irrigation systems, the complexity of PV modules, inverters, and battery configurations makes digital operation and maintenance more urgent.

From a broader perspective, digital twins and dynamic models are poised to become standard components of advanced rural energy systems. Especially in conservation agriculture, mainstream approaches are transitioning from static estimation to dynamic thermal modeling [53]. Future systems are expected to further integrate real-time meteorological data, crop growth models, and energy price signals, thereby achieving closed-loop optimization of energy utilization, agricultural yield, and economic benefits.

(5) Policy and Market Transformation

Large-scale deployment of distributed rural energy requires strong institutional guarantees. The policy framework for rural distributed energy is evolving toward more flexible self-consumption, shared generation, storage participation, and multi-stakeholder electricity trading mechanisms [46]. Approval procedures, capacity thresholds, grid-connection rules, load-management requirements, and settlement mechanisms directly influence the revenue structure and investment feasibility of rural energy projects. The electrification of agricultural machinery, including electric tractors, also depends on coordinated policy incentives, because equipment replacement involves high upfront costs, uncertain battery-life expenses, and operational-performance concerns. Therefore, targeted agricultural policy funds and market-oriented incentives are needed to support machinery renewal and to form a positive interaction between technology demand and industrial supply.

At present, there is a zero-sum conflict between PV shading and crop yield in the agrophotovoltaic complementary system [50], and the sensitive economic model and complex operation and maintenance costs make the situation worse [53]. In the foreseeable future, it will be necessary for energy systems to transition from passive layout to active light management, with the support of translucent photovoltaic modules (e.g., OPV) [49]. There is also the core trend of constructing PV thermal coupling and multi-energy synergies, such as heat pumps, to reduce the dependence on electrochemical batteries and realize the real collaborative optimization of agriculture and energy [56].

2.3. Overview of Smart Agricultural Technology

Table 3 summarizes the smart agricultural technology studies reviewed in this subsection, including climate-smart agriculture, prediction and irrigation, machine vision, agricultural IoT, communication and edge infrastructure, cybersecurity, blockchain traceability, and technology diffusion.

Table 3. Representative studies on smart energy technologies.

Ref	Year Range	Research Emphasis	Key Findings
[63–65]	2022–2025	Climate-intelligent agriculture and improvement of climate resilience	Multi-tech synergy enhances yield, adaptation, and regional smart agriculture
[66–68]	2023–2025	Farmland environment prediction, intelligent irrigation, and water management	Enhanced agri-meteorology prediction supporting precision irrigation
[69–72]	2022–2025	Intelligent perception, machine learning, and agricultural visual recognition	Practical agricultural AI via deep learning and edge computing
[73–75]	2024–2025	Agricultural Internet of Things, intelligent sensors, and greenhouse monitoring and control	IoT-enabled sensing for environmental monitoring and agri-energy management
[76–78]	2022–2025	Communication, positioning, and edge computing infrastructure	Infrastructure support for smart agri: machinery, monitoring, real-time operations
[79,80]	2023–2024	Agricultural network security, system protection, and digital forensics	Smart agriculture security: multi-layer threats, protection strategies, and digital forensics
[81]	2024	Blockchain traceability and credible agricultural product supply chain	Blockchain and IoT for trusted agri-supply chains: transparency and sustainability
[82]	2022	Evaluation, extension, and standardized application of smart agricultural Technology	Data-driven support for large-scale smart agri tech adoption

2.3.1. Current Limitations of Smart Agriculture Technology

(1) Sensing and Data Acquisition Constraints

Smart agriculture heavily relies on sensors to collect critical parameters like temperature, soil moisture, nutrient status, and radiation. However, the complexity of agricultural environments introduces multiple sources of error and frequent data loss issues. Sensor failures, battery depletion, electromagnetic interference, extreme weather, and equipment damage can all lead to distorted measurement data [63–65]. In addition, sensors are often subjected to accidental displacement or damage due to human factors (e.g., personnel movement), animal activity, or collisions with agricultural machinery, resulting in data corruption or loss, which directly undermines the availability and reliability of the data.

These limitations are particularly critical in closed-loop control scenarios such as irrigation. Due to the nonlinear and non-monotonic multivariate relationships, significant outliers, and complex interaction effects in agricultural systems, noise or incomplete data can substantially amplify decision-making risks [66–68]. Consequently, under actual field conditions, control models may exhibit misjudgments and operational instability.

(2) Network and System Security Risks

A typical smart agricultural Internet of Things framework is structured into four core tiers: the sensing layer, network layer, cloud platform layer, and application layer. Within this architecture, the network layer must simultaneously fulfill two critical requirements: supporting large-scale upstream monitoring data transmission while ensuring reliable downlink transmission of control commands. Yet during practical deployment, this tier faces significant exposure to cyber threats such as DoS, DDoS attacks, routing tampering,

wiretapping, and man-in-the-middle intrusions [69–72]. These threats may cause communication delays, packet loss, service interruptions, and even delayed activation of critical equipment like pump valves.

The more fundamental challenge lies in the absence of standardized agricultural IoT architectures and security baselines. This deficiency not only weakens interoperability between devices and platforms but also impedes the establishment of unified security protection and audit frameworks [73–75]. As agricultural production increasingly relies on digital systems, cybersecurity incidents can not only cause direct production losses but also trigger broader economic and social ripple effects.

(3) Algorithm and Modeling Challenges

While many smart agriculture models demonstrate strong performance in laboratory or short-term experimental settings, their practical applicability in long-term field operations remains questionable. A key challenge lies in their limited generalization capacity across regions and seasons. Agricultural production conditions vary significantly across different areas, climate types, crop varieties, and farming practices [76–78]. This inherent heterogeneity often results in a sharp decline in model performance when applied to new locations, crops, or seasonal transitions.

Another critical limitation lies in the disconnect between model evaluation cycles and actual agricultural production cycles. Current research predominantly relies on laboratory tests or short-term trials, failing to track model performance over extended periods across diverse climates, soils, and cultivation systems. Consequently, the robustness and enduring practical value of these models remain challenging to assess comprehensively. Simultaneously, the opaque nature of numerous advanced models further hinders their practical deployment. Since agricultural decisions exert a direct influence on farmers' livelihoods and food security, end users demand not only accurate recommendations but also interpretable reasoning behind every decision. When black-box models fail, it often proves difficult to determine whether the root cause stems from insufficient data, improper transfer, or inherent flaws in the model structure, which undermines user trust.

In visual intelligence tasks such as fruit detection and yield estimation, these challenges are exacerbated by specific scene factors, including object occlusion, cluttered backgrounds, repeated counting, and difficulties in identifying small objects. These issues significantly reduce detection reliability and highlight the broader challenges faced in deploying visual perception systems in complex agricultural environments.

(4) Safety and Accountability Gaps

Unlike system availability and functional deployment, security, privacy, and accountability issues in agricultural digital systems have long been neglected. Weak cybersecurity protections and inconsistent IoT architectures in agriculture make smart farming systems vulnerable to cybercrimes and malicious attacks [79]. Moreover, the absence of standardized forensic frameworks for agricultural IoT complicates incident investigations, traceability, liability determination, and post-disaster recovery [80]. As a result, existing systems often lack the controllability and resilience required for long-term reliable deployment.

(5) Deployment Constraints

The field deployment of smart agricultural technologies is typically conducted under highly constrained conditions, including unstable power supply, limited network coverage, and insufficient operation and maintenance (O&M), as well as technical support. These challenges are particularly severe in developing countries [81]. In this context, the high training and deployment costs of agricultural foundational models and multimodal models fundamentally conflict with the low-profit economic structure of agriculture.

Despite the introduction of technical solutions such as model compression, knowledge distillation, quantization, and edge computing, significant challenges remain. The noise in agricultural data and the complexity of real-world scenarios can reduce the accuracy and stability of compressed models. In addition, the reliability of edge hardware and its robust offline operation capability under harsh field conditions remains difficult to guarantee.

(6) Technology Diffusion Barriers

This demonstrates that technical complexity alone is insufficient to ensure scalable deployment. To enable smart agricultural technologies to extend beyond pilot-scale implementation, factors such as maintainability, training requirements, infrastructure compatibility, and local service capabilities must be incorporated into system design and evaluation [82].

2.3.2. Development Trends of Smart Agriculture Technology

The progress of intelligent agriculture technology is gradually shifting from isolated perception and task-specific optimization to integrated, flexible, and knowledge-driven system architecture. A prominent trend is the transition from single sensor deployment to system-level coordination based on multi-source data fusion and device edge cloud collaboration. In order to cope with the persistent challenges of environmental noise, target occlusion, signal interference, and information fragmentation, the future intelligent agricultural system will more widely use multi-sensor and multi-modal fusion strategies. These strategies may include the combination of vision, sonar, and radio frequency identification (RFID). This integration can improve the robustness and accuracy of monitoring and positioning, and meet the requirements of low power consumption, cost control, and the reliability of the whole system. In irrigation and other control-oriented scenarios, decision-making is also moving beyond simple threshold-based control to a unified framework that considers soil moisture, temperature, radiation, and nutrient status. Therefore, more refined and adaptive regulation strategies are expected to improve the utilization efficiency of water and energy [83], while location-tailored control methods will play an ever more critical role in addressing the marked spatial variability of agricultural settings.

At the communication level, intelligent agriculture is developing in the direction of wider coverage, lower power consumption, lower deployment costs, and stronger offline operation ability. The future agricultural communication system will pay more attention to low-power wide area connection, wide coverage, and anti-interference ability, and achieve a higher degree of autonomy at the equipment level. Technologies such as Low-Power Wide-Area Networks (LPWAN) and long-range wireless communication have shown strong potential in applications including microclimate monitoring and irrigation demand estimation, particularly because they reduce dependence on continuous and high-quality network access. At the same time, device-side data caching, breakpoint recovery, and edge-side fault-tolerant control are likely to become standard system functions, especially in rural areas where network instability and unreliable power supply remain common [84]. These capabilities are essential for maintaining functional continuity under conditions of limited infrastructure.

The other main direction is a multimodal basic model and a domain-specific agricultural model. This may accelerate the transition from task-oriented intelligence to integrated perception, knowledge, and temporal reasoning to a unified intelligent backbone. The rapid development of multimodal models is expected to reshape smart agriculture by supporting more integrated solutions for crop phenotypic analysis, environmental monitoring and early warning, intelligent question answering, and agricultural knowledge graph construction. By combining images, remote sensing data, sensor time series, and textual knowledge, such models can substantially improve the capacity of agricultural systems to address complex multi-source problems [84]. However, future progress is unlikely to

depend solely on increasing model scale. Instead, research will increasingly focus on three aspects: improving multimodal representation and unified embedding spaces for stronger cross-modal reasoning; enhancing deployment feasibility under constraints of cost, computational resources, and energy supply through model compression, distillation, and edge inference [84]; and incorporating long-term field validation together with cross-seasonal evaluation into the model lifecycle so that laboratory performance can be better aligned with real agricultural value.

As intelligent systems become more deeply involved in agricultural decision-making, trustworthy AI and interpretable decision-support mechanisms will become indispensable. Because agricultural decisions directly affect production outcomes, farmers' livelihoods, and food security, future Smart Agricultural systems must move beyond simply generating answers or recommendations. Instead, they should provide interpretable evidence, explicit uncertainty information, and clearly defined risk boundaries. Embedding interpretability, auditability, and error-diagnosis mechanisms into both models and decision-making processes can help reduce resistance to black-box systems and improve user trust [84]. This trend also implies a more consultative mode of decision-making, in which model outputs are not treated as unquestionable instructions but are assessed jointly with farmers' experiential knowledge [84]. Such a human-centered interactive framework is likely to become increasingly important in real-world practice.

At the same time, security, privacy protection, and digital forensics are shifting from passive defense to system designs that are more resilient, traceable, and recoverable. Agricultural IoT architectures are often non-uniform, and their security baselines remain weak. In response, future research is expected to focus on layered security mechanisms. It is also likely to emphasize encryption of critical communication links, anomaly detection, rapid response systems, and digital forensic frameworks tailored to agricultural scenarios [79]. This indicates a clear shift from passive protection toward resilient architectures capable of traceability, timely response, and post-incident recovery. In parallel, to mitigate common network-layer threats such as distributed denial-of-service attacks, routing tampering, and eavesdropping, future systems will likely adopt stricter access control, stronger key management, intrusion detection, graceful service degradation [80], and business continuity design to ensure the availability of core agricultural functions.

Hardware innovation will also remain a foundational driver of Smart Agriculture, with development advancing toward higher reliability, lower power consumption, and stronger environmental adaptability. Breakthroughs in advanced materials and micro-/nano-scale devices are expected to enable more reliable and energy-efficient long-term deployment in agricultural environments. For example, optimized gallium nitride (GaN) barrier microelectromechanical systems (MEMS) show clear advantages in temperature adaptability, sensitivity, and operational stability. These features make them particularly suitable for agricultural applications. This is especially true for scenarios that require tolerance to extreme temperatures and ultra-low power consumption [75]. These developments suggest that hardware progress will continue to underpin the expansion of smart agriculture, especially in harsh environments where sensors must satisfy exceptionally demanding requirements for durability and energy efficiency.

Finally, productization and service-oriented development will be crucial for large-scale adoption of Smart Agricultural technologies. To overcome longstanding barriers such as high initial investment, inadequate infrastructure, difficulties in data collection, insufficient operational expertise, and uncertain returns, future smart agriculture solutions will need to place greater emphasis on modular product design. They will also need to support plug-and-play deployment, low-maintenance operation, and simplified training and promotion [82]. Such features are essential for reducing user burden and improving

the feasibility of broad implementation. At the same time, demonstration projects and service ecosystems will play an increasingly important role in raising technological visibility, strengthening farmers' trust [82], and building the institutional as well as technical support systems needed for sustained dissemination and adoption.

The result of smart agriculture is shifting towards an integrated, knowledge-driven system-level coordination architecture. By using multi-source data fusion, such as combining vision, sonar, radio frequency identification, etc., this system can enhance the robustness of noise monitoring in harsh environments [85]. Comprehensive perception enables a shift from simple threshold triggering to a unified framework considering soil, temperature, radiation, etc., which can realize precise and region-specific control strategies and greatly improve the utilization efficiency of water and energy [83]. In order to support architectures with a large amount of data in rural areas with weak power grids, communication technologies such as low-power wide area network (LPWAN) are widely applied to reduce the dependence on high-quality network access [86]. Storing data at the device end, being able to resume from breakpoints, and having edge end fault-tolerant control are standard system functions to ensure that offline operations can be carried out independently [84].

A calculation-centered multimodal foundation model combines images, remote sensing, sensor time series, text, etc., to solve agricultural problems. To overcome the challenges in rural deployment, research resources focus on model compression, knowledge distillation, edge inference, and long-term cross-season field verification, so as to match the laboratory performance with agricultural value. However, with the deep involvement of artificial intelligence systems in key agricultural decisions that affect farmers' livelihoods, "trusted artificial intelligence" is key. Place interpretability, auditability, and error diagnosis mechanisms in the decision-making process, reduce user dependence on "black-box" systems, and create a mode to evaluate model outputs together with farmers' empirical knowledge.

To ensure the long-term sustainability of the digital system, agri cybersecurity shifts from passive defense to a resilient layered mechanism. Emphasizing encryption, anomaly detection, digital forensics framework [79] and graceful service degradation [80] is critical to mitigate network-layer threats (e.g., DDoS, route tampering) and ensure core functions continue to be available. Key hardware innovations supporting these digital and security advancements, such as optimized GaN MEMS, provide ultra-wide temperature adaptability and ultra-low power consumption for harsh field deployments [75]. Finally, to cross the threshold of large-scale adoption, it is necessary to rely on modular product design, plug-and-play deployment, and a solid service ecosystem (such as demonstration projects, etc.). Such operations can lighten the burden for users and build up the support of the required institutions [82].

The traditional agricultural Internet of Things has high environmental noise [83] and relies on the cloud, so it is very "more perception and weaker execution", the system is relatively fragile, and real-time control cannot be realized in rural areas with weak power grids [80]. In addition, this bottleneck requires migrating computing power to the edge (edge artificial intelligence); constructing a local low-latency control loop, building an artificial intelligence model, and enhancing network security, which are relatively important for the continuity of functions and the intelligent resilience of agriculture [84].

The key conclusions are in an isolated state in view of the gaps in the comprehensive literature: the isolated technical path has reached the limit of theoretical practice. However, the progress of agricultural electrification is constrained by elevated energy storage costs as well as temporal and spatial mismatches. Agricultural informatization is troubled by unstable rural networks and the lack of controllable physical actuators. To systematically overcome the bottlenecks, the agricultural process needs to be redefined as a schedulable

physical load predicted by advanced artificial intelligence and managed by the edge computing brain. This paradigm shift directly requires the introduction of the AEI.

3. Theoretical Evolution and Technology Stack of AEI

3.1. Overview of the Technology Evolution Roadmap

Table 4 summarizes the stage-based evolution of AEI-related technologies before the detailed narrative discussion, so that the first appearances of the cited studies follow numerical order within this subsection.

Table 4. AEI technology evolution roadmap.

Phase (Years)	Research Paradigm/Core Objectives	Main Research Areas	Key Research Questions	Key Technologies	Main Functions	Ref
Early Development of Digitalization and Precision Agriculture (2000–2009)	Digitalization of farmland/produce helps precision operation	On-site information modeling, prescription mapping, first-generation monitoring	Data sparsity, high sensor costs, and lack of unified standards	GPS/GIS, early perception and data recording, prescription map-driven variable speed operation	Using GIS+ sampling/statistical model to generate prescription map, realize variable fertilization/pesticide spraying, and other “semi-automated decision-making	[87]
Agricultural Internet of Things + Cloud Platform (2010–2015)	Connect everything and make it visible, creating a remote monitoring and farm management platform	Ag-IoT Architecture, Device Connectivity, Cloud-Based Farm Management, and Traceability	Poor interoperability, heavy reliance on cloud services, high energy consumption of devices, and heavy maintenance burden	RFID/Barcode traceability technology, ZigBee/802.15.4 standards, cloud platform, and network services	The “sensor-gateway-cloud platform” enables remote dashboard, alarm, and device coordination functions; the traceability system correlates logistics with identity.	[88–90]
The Introduction of Big Data and Edge/Fog Computing (2016–2019)	“More efficient computing, more real-time capability” to alleviate rural network limitations and cloud latency	Edge-cloud collaboration, streaming processing, LPWAN, multi-source fusion (ground + drones/remote sensing)	Low latency, offline availability, and cost-effective transmission of massive data	Edge/fog computing; edge-cloud platform; LP-WAN/LoRaWAN performance modeling	The gateway/edge node performs preprocessing and local decision-making (thresholds, lightweight models), uploading only “features/event”; closed-loop control can still be achieved during power outages	[91,92]
Artificial Intelligence-driven + Trusted Collaboration + CPS Formation (2020–2022)	“Understanding, Ability, and Trust” to Build Agricultural CPS	Deep learning perception (disease/growth/environment), digital twin, blockchain-based trusted traceability, security, and privacy	Weak model generalization capability, data silos, privacy compliance, and system security	Agricultural Digital Twin Technology Roadmap; Blockchain + Edge Collaboration	Utilize deep learning (DL) for visual/time series recognition; employ digital twin technology for hypothesis simulation and strategy validation; adopt blockchain technology to ensure tamper-proof traceability	[93–95]
Agriculture-Energy-Environment Synergy Optimization (2023–2026)	Cross-domain collaborative optimization: unified revenue/cost/carbon/reliability	Agricultural Microgrid/Rural Energy Internet, Controllable Greenhouse Load, VPP, Carbon Accounting and MRV, Coupled Energy-Information Security	Multi-objective coupling (crop-energy-carbon), uncertainty (weather/load/price), market mechanism, and dispatching	Key Technologies of the Agricultural Energy Internet (INPA); Overview of Microgrid EMS; Carbon-Constrained Microgrid Dispatching; Review of VPP	Joint dispatch of “controllable agricultural loads (greenhouse environment/auxiliary lighting/irrigation) + distributed energy resources (DER, PV/storage/biomass)” with carbon accounting/MRV incorporated as constraints or targets in closed-loop optimization	[96,97]

The evolution of agricultural Internet technology: from agricultural digitization and the Internet of Things (IoT) to edge intelligence, trustworthy collaboration, and then to the integrated optimization of agriculture—energy—environment. Each transformation corresponds to a system upgrade that solves the previous limitations. The first stage (2000–2009) focused on digitization and precision agriculture, using geographic information systems (GIS) and global positioning systems (GPS) to provide semi-automatic decision-making support for variable-rate operations [87]. However, this stage had challenges like scattered data sources and high costs. To solve the problem, the second stage (2010–2015) introduced agricultural IoT and cloud platformization. Adopt RFID tracing, short-distance communication such as Zigbee, and cloud architecture [90], and this stage realizes end-to-end remote monitoring. However, this cloud-dependent model is prone to interoperability problems, high equipment energy consumption, and also has reliability risks related to unstable rural networks.

There are problems of transmission delay and network security restrictions in the 2016–2019 stage. By adopting edge and fog computing and integrating technologies such as LPWAN, there is a hybrid paradigm with edge local preprocessing and so on [91], which can ensure the function of core agricultural control when the network is disconnected. The 2020–2022 stage has entered the era of artificial intelligence-driven, reliable collaboration and agricultural CPS. Deep learning has enabled an increase in environmental awareness. Digital twin technology transforms systems from basic monitoring to a verifiable and predictable decision-making platform [93]. Blockchain helps with cross-entity data sharing and traceable auditing [95], reducing data silos and trust issues in multi-party collaborative operations.

Finally, to solve the problems of space-time energy mismatch and economic issues in Sections 2.1 and 2.2, the fifth stage (2023–2026) proposes an AEI paradigm. Under the dual-carbon goal, this stage will study the expansion from an isolated agricultural production network to a unified framework, which includes controllable agricultural loads, distributed energy resources, and market mechanisms. The core scientific challenges have changed, and it is necessary to achieve coordinated optimization among agriculture, energy, and the environment under the constraints of weather, load, price, and crop physiology. Multi-energy optimization, virtual power plant aggregation, and predictive closed-loop control, AEI turns the agricultural process into a dispatchable energy asset. This transformation from single-point technical value to cross-field collaborative energy management provides the basic theoretical basis for the power coupling of deep information in Section 4.

The technical evolution from isolated agricultural Internet of Things to AEI has provided a clear roadmap for solving rural network delays and energy time-space mismatches. Integrate edge computing, digital twin, and cyber-physical system, construct a unified framework to coordinate and optimize the physical process of agriculture and the resources of distributed energy, and create the coupling of the main lines of agricultural informatization and rural electrification. The overall development trajectory is summarized in Figure 2.

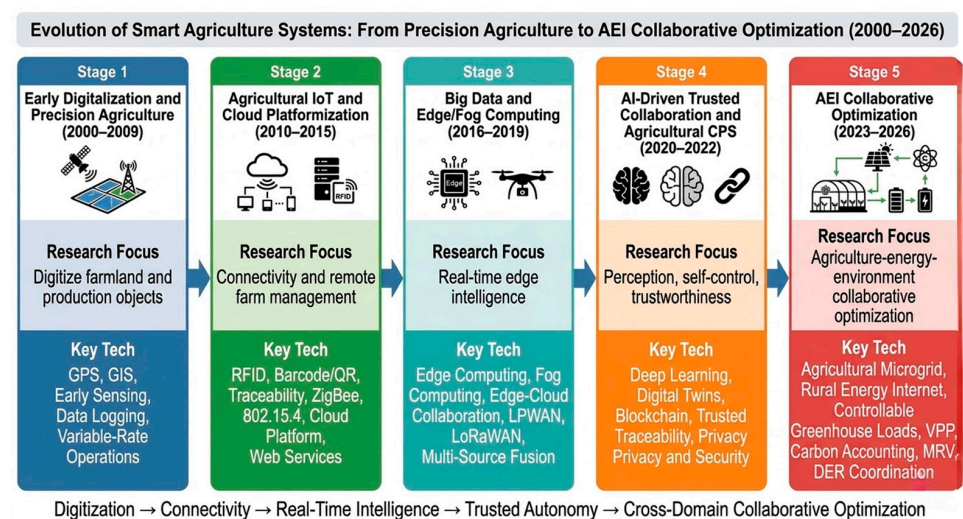


Figure 2. AEI Development Roadmap.

3.2. Technical Framework

The technical framework of the AEI is not limited to a single platform or specific communication network, but rather encompasses an end-to-end technology chain that includes physical entities, multi-source sensing, interoperability, edge control, data governance, digital twins, artificial intelligence, optimization algorithms, and application decision-making.

Meanwhile, security and trustworthiness serve as cross-cutting capabilities that permeate all technical layers, ensuring data availability, controllability, and auditability in cross-entity collaboration. The layered structure of this end-to-end AEI technology chain is illustrated in Figure 3.

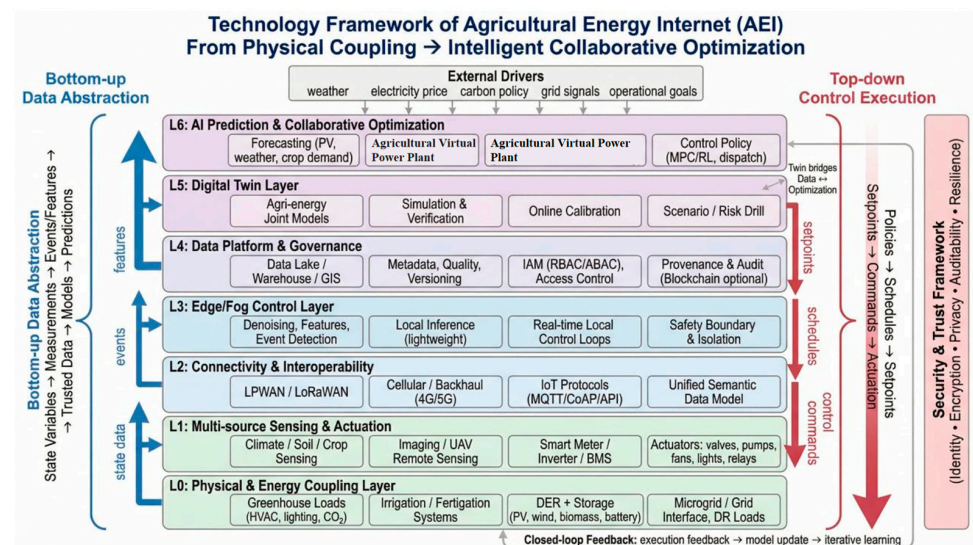


Figure 3. AEI Technical Framework.

3.2.1. Physical Entity and Energy Coupling Layer

The uniqueness of the AEI system is primarily reflected in its expanded and integrated regulatory domains. Unlike traditional agricultural internet systems that focus on production entities like farmland and greenhouses, AEI must simultaneously integrate both agricultural production and energy systems within its framework, enabling unified modeling and coordinated regulation of production variables and energy processes. Key regulatory components on the agricultural side include greenhouse environmental systems (heating, ventilation, supplementary lighting, CO₂ enrichment), irrigation and drip irrigation systems, as well as aeration/circulation systems for livestock and aquaculture. On the energy side, core regulatory elements encompass distributed renewable energy sources (PV, wind turbines, biomass), energy storage systems, microgrid-distribution network interfaces, and controllable loads with demand response capabilities. This hierarchical structure defines AEI's core state variables (e.g., environmental parameters, crop growth metrics, load power) and establishes system constraints (e.g., optimal crop growth ranges, grid operational limits, energy storage lifespan, and power ceilings). These elements form the foundation for the upper-level data platform and intelligent decision-making execution.

Based on this, energy systems typically establish a pool of dispatchable resources through microgrid energy management systems. The formulation of optimization problems aims to achieve economic benefits, reliability, and carbon emission constraints. The power and energy sectors have developed a relatively mature decision-making modeling research framework for microgrids, encompassing objective functions (cost, carbon emissions, equipment lifespan), constraints, and solutions (deterministic, stochastic, robust, distributed, etc.) [98].

The new challenge for AEI lies in the fact that the traditional agricultural load is not a rigid load, as its power characteristics are highly coupled with crop growth stages and environmental control strategies. Therefore, it is necessary to integrate agricultural machinery models, empirical models, or data-driven models into the energy management process to achieve synergistic objectives in energy, environment, yield, and quality.

3.2.2. Multi-Source Perception and Driving Layer

Without a stable and calibrated multi-source sensing and drive chain, the AEI would degenerate into a mere energy consumption statistics platform, incapable of achieving closed-loop optimization. The core mission of this layer is to establish a unified observation system for both agricultural and energy sectors. In agriculture, it primarily involves meteorological and microclimate sensing (temperature, humidity, CO₂ concentration, light, VPD), soil moisture and nutrient sensing (EC, pH), as well as pest and crop phenotyping sensing (multispectral/visible light imaging, drone remote sensing, etc.). In energy, it focuses on smart meters, inverters, battery management systems, protection/switching status monitoring, and power quality monitoring. The drive side encompasses valves, pumps, fans, shading systems, auxiliary lighting, heating/cooling equipment, controllable sockets/relays, as well as energy storage charging/discharging and interruptible/transferrable load control devices.

The core challenges at this level are mainly reflected in two aspects: equipment heterogeneity and calibration/operation costs, as well as the temporal and spatial inconsistency of sensor data and noise problems. For example, the change of sensor arrangement in a greenhouse may lead to the deviation of on-site temperature and humidity monitoring, so that the agricultural imaging data are significantly affected by light fluctuations. Existing reviews have always emphasized that the deployment of sensor networks, energy supply (including power collection), and long-term reliability are still the key bottlenecks to achieve sustainable system operation under the agricultural scenario. Therefore, within the framework of AEI, sensor maintainability and actuator controllability should be paid equal attention to as algorithms.

3.2.3. Interconnection and Interoperability Layer

In the traditional agricultural Internet system, the basic connectivity is enough to realize remote monitoring. But AEI needs closed-loop control and cross-entity collaborative scheduling. This imposes higher technical standards on the interconnection layer in terms of reliability, latency optimization, broad coverage, and cross-system compatibility. In rural areas, low-energy WAN has become the key technology of Agri-IoT with its advantages of low power consumption and long-distance coverage. Intelligent agriculture performance modeling and empirical research established a clear relationship between node size, transmission interval, and data transmission success rate, providing quantitative guidance for network planning of large-scale farmland and scattered sites [99]. Simultaneously, communication link quality exhibits more pronounced fluctuations in mobile agricultural nodes such as machinery, robots, and drones. Recent studies have proposed a LoRa adaptive rate enhancement algorithm tailored for agricultural mobile scenarios [100], which dynamically adjusts parameters based on channel characteristics like received signal strength indicator (RSSI) and signal-to-noise ratio (SNR) to improve communication robustness.

At the protocol level, mainstream IoT messaging protocols (e.g., MQTT and CoAP) can be used to build lightweight data channels. However, when interfacing with energy systems, it is essential to ensure consistency with power-side systems (e.g., microgrid control and distribution network automation systems) in terms of data models and interface specifications, to prevent the formation of new data silos between agricultural platforms and energy-side EMS/SCADA systems. Therefore, the interconnection layer of AEI recommends adopting a hierarchical protocol stack architecture, combined with unified data models and semantic specifications. The lower layer supports multi-network coexistence, while the upper layer achieves integration of agricultural and energy data flows through unified device abstraction and topic/event models.

3.2.4. Edge/Mist Computing Control Layer

The third phase's pivotal transition in the technology evolution roadmap (Section 3.1) involves the adoption of edge/fog computing. Its value lies not only in reducing cloud traffic but more importantly in enabling low-latency, offline-accessible local autonomy. As reviewed in the agricultural internet domain, edge computing provides a practical computational model for smart agriculture, particularly suitable for scenarios with limited network connectivity, dispersed device distribution, and rapid response requirements [100]. In AEI, the edge layer usually performs three functions: (a) the data plane cleans data, reduces noise, extracts features, detects events, caches information, and uploads only features or exceptions when bandwidth is limited; (b) the control plane manages local control loops for greenhouse environments, irrigation, and energy-storage systems to ensure safe operation during network interruption; and (c) the collaboration plane reduces the exposure of cross-entity collaboration by implementing device authentication, access control, and area isolation as security boundaries.

Moreover, investigations into edge-cloud collaboration platforms reveal that developing the agricultural IoT platform based on an edge-cloud collaborative architecture allows it to reconcile global optimization with real-time performance, alongside the rapid response and localized optimization brought by edge processing [91], while the cloud manages global optimization, model training, and cross-domain collaboration.

This division of labor is essential for AEI. Although coordinated agricultural and energy scheduling usually requires global prediction and optimization, for ensuring system safety and real-time responsiveness, actual execution has to be conducted on the edge side.

3.2.5. Data Platform and Governance Layer

As the participants of AEI expand from a single farm to cooperatives, industrial parks, power grid operators, and market entities, its core data challenges have evolved from “data availability” to “data credibility”, “cross-domain sharing capability”, and “post sharing audit and traceability mechanism”. It can be seen that AEI's data platform must not only provide traditional functions such as data lake, data warehouse, metadata management, and GIS spatio-temporal index, but also implement trust mechanisms and governance processes, including identity and access management, data quality control, version control, and data traceability tracking.

In terms of trust mechanisms, blockchain technology has been widely applied in agricultural supply chain traceability and multi-party collaboration scenarios. A comprehensive review system has systematically analyzed the development trends and challenges of blockchain in agricultural traceability, highlighting its potential in tamper-proof auditing and multi-party consistency assurance. However, it also points out that system throughput, privacy protection, and on-chain/off-chain coordination remain challenges in engineering [101]. For the AEI, the value of such mechanisms extends far beyond agricultural product traceability—they can also provide an auditable and credible ledger foundation for carbon accounting, demand response settlement, and cross-entity collaborative scheduling.

3.2.6. Digital Twin Layer

Within Phase IV of the technology development roadmap (detailed in Section 3.1), a key advance lay in the adoption of digital twin techniques, upgrading the system from fundamental monitoring and regulation to an intelligent framework capable of simulation, validation, and forecasting. The study for agricultural scenarios proposed an application roadmap for digital twin technology, summarizing its application scenarios and value in the agricultural sector between 2017 and 2020 [93]. It demonstrated that this technology

could effectively enhance equipment maintenance, process optimization, and decision support capabilities.

Within the AEI framework, digital twin technology has the following three key functions: (A) consistent modeling: integrating the dynamic characteristics of greenhouse environmental control, irrigation and fertilization, distributed energy and storage systems into a unified simulation environment, and establishing a joint model across domains. (B) Strategy simulation and verification: test various scheduling strategies without interrupting the actual production (for example, combine the peak valley electricity price with the lighting strategy, and adjust the irrigation timing under the carbon constraint), so as to effectively reduce the trial-and-error cost. (C) Online calibration and risk drill: use real-time data to calibrate model parameters and simulate scenarios such as extreme weather, equipment failure, and price fluctuation to enhance the risk elasticity of the system.

For energy-production systems, the digital twin should include renewable-generation forecasting, converter and storage status, grid-connection constraints, energy balance, thermal-load balance, and agricultural demand forecasting. By coupling these elements with greenhouse, irrigation, livestock, and processing models, the twin can test energy-management strategies before field deployment and can update the dispatch plan when weather, prices, or production stages change.

As a result, the digital twin layer essentially functions as a connection between the data platform and optimization decisions, ensuring that the AI's intelligent decisions are supported by interpretable simulation results rather than relying solely on black-box outputs.

3.2.7. Artificial Intelligence and Collaborative Optimization Layer

The intelligent optimization layer mainly completes the following two core functions: one is uncertainty prediction and state estimation, and the other is multi-objective collaborative optimization and closed-loop control.

- (1) Prediction and Estimation: Both agricultural and energy systems face significant uncertainties.

For instance, weather conditions affect PV power generation and greenhouse heat loads, crop growth stages alter light and water requirements, and market prices influence energy dispatch strategies. Therefore, Smart Agricultural energy systems typically require multi-scale prediction frameworks covering minute-level control, hourly scheduling, and daily/weekly planning. Lightweight online inference is implemented at edge devices, while more complex model training and calibration are completed in the cloud. These prediction tasks not only involve traditional time series models but also incorporate deep learning models like LSTM and Transformer, along with probabilistic prediction methods, supporting stochastic or robust optimization.

- (2) Collaborative Optimization and Control: MEMS research in the power and energy sector provides uncertainty optimization paradigms (stochastic, robust, opportunity constrained programming [98]) for AEI energy dispatching.

The key progress of AEI is that it integrates controllable agricultural loads into decision variables. As an example, it is feasible to adjust the scheduling of greenhouse lighting and heating, which can balance the peak and valley power demands while meeting the physiological requirements of crops.; optimizing irrigation plans to coordinate with electricity pricing or PV output under soil moisture threshold constraints; and explicitly integrating emission factors and carbon costs into dispatch objectives within carbon constraints or trading mechanisms.

When AEIs participate in market transactions through entities such as industrial parks or cooperatives, they typically utilize Virtual Power Plants (VPPs) for data aggregation,

bidding, and settlement. The review systematically categorizes VPP models and their interactions with power markets [102], covering modeling methodologies, solutions, and market participant types, thereby providing reusable market-level design concepts for the large-scale commercial operation of AEIs.

3.2.8. Quantitative Evaluation Indicators and Objective Functions

For AEI research, the optimization objectives and evaluation criteria should be clearly stated. The general multi-objective optimization expression can be expressed as:

$$\text{Min } J = w_1 \times C_{\text{op}} + w_2 \times E_{\text{carbon}} + w_3 \times L_{\text{unserved}} + w_4 \times Q_{\text{power}} + w_5 \times R_{\text{safety}} - w_6 \times P_{\text{agri}} \quad (1)$$

where C_{op} denotes operating cost, E_{carbon} denotes carbon-emission cost or penalty, L_{unserved} denotes unserved energy or outage loss, Q_{power} denotes power-quality deviation, R_{safety} denotes electrical safety risk, and P_{agri} denotes agricultural production benefit. Specific items can be adjusted according to different scenarios, but the objective function structure should always clearly reflect the trade-off relationship between various factors.

In this expression, the weights w_1 – w_6 should be determined according to scenario priorities rather than treated as abstract parameters. For instance, greenhouse and cold-chain scenarios should assign higher weights to outage loss and safety risk, whereas park-level energy management may assign higher weights to carbon cost and renewable-energy utilization. Constraints should also be explicitly stated, including power balance, thermal balance, equipment capacity, state-of-charge limits, crop/livestock physiological thresholds, voltage/frequency limits, and maximum allowable interruption time.

Correspondingly, this review emphasizes a set of universal quantitative indicators: energy self-sufficiency rate, renewable energy utilization rate, annualized cost or cost per kilowatt hour (LCOE), unit energy intensity (such as electricity consumption per ton of product, electricity consumption per square meter of greenhouse area, or electricity consumption per cubic meter of irrigation water), power outage probability/reliability indicators, voltage deviation, frequency deviation, total harmonic distortion rate, outage tolerance time, product loss or downtime cost, carbon intensity, and crop/livestock production performance indicators. These indicators make comparisons between studies beyond descriptive discussions, and are more comparable.

To make these numerical characteristics explicit, Table 5 summarizes the recommended quantitative indicators for AEI evaluation and links them to the review criteria.

Table 5. Quantitative Evaluation Framework for Agricultural Energy Internet Systems.

Evaluation Dimension	Representative Numerical Indicators	Purpose in AEI Analysis
Energy and load	daily/seasonal electric-load curve; heat-load curve; kWh/t product; kWh/m ² greenhouse; kWh/m ³ irrigation water	Identify process energy intensity and seasonal mismatch
Reliability and power quality	SAIDI/SAIFI or outage probability; unserved energy; allowable outage duration; voltage deviation; frequency deviation; THD	Assess supply dependence, power quality, and electrical safety
Agricultural production	yield, quality grade, livestock survival/health index, cold-chain product-loss rate, microclimate violation duration	Link energy control to production outcomes
Environment and carbon	kg CO ₂ ^{−eq} /t product; renewable utilization rate; life-cycle equipment burden; waste-to-energy conversion rate	Evaluate low-carbon performance beyond farm operation
Economics and optimization	annualized cost, LCOE, payback period, carbon cost, downtime cost, net benefit	Clarify objective functions and trade-offs

3.2.9. Application and Decision-Making Level

Based on the aforementioned technology chain, the application layer of AEI typically manifests as “decision products” for industrial stakeholders to comprehend and utilize. This encompasses four categories of decision-making content:

- (1) Production decision-making: environmental parameter setting strategy, integrated irrigation and fertilization implementation plan, pest and disease early warning and control recommendations.
- (2) Energy-side decision-making: energy storage charging/discharging schedules, demand response implementation plans, and microgrid grid-connection/islanding strategies.
- (3) Environmental decision-making: carbon accounting and emission reduction pathway planning, carbon asset management, and MRV documentation.
- (4) Operational decisions: park-level coordinated dispatching plans, VPP market participation strategies, and settlement and audit arrangements.

It is crucial to emphasize that AEI applications should not be limited to data visualization, but should establish a closed-loop operational model encompassing “strategy formulation → execution → feedback → iterative learning”. Execution and feedback data from the edge will be uploaded to the platform to calibrate and update the digital twin model, thereby driving continuous optimization of the strategic system and enabling dynamic iteration throughout the process.

3.2.10. Horizontal Support System

The cross-domain coupling characteristics and multi-party collaboration model of AEI need to have strict system security and trust. Abnormalities in agricultural production control will cause economic losses, and power grid interaction needs to be strictly compliant, so security is the horizontal constraint of the entire technical system. This covers equipment-side authentication, network-side encrypted transmission, platform-side data management, and algorithm-side privacy protection, and needs to be fully covered at all layers. Research into blockchain-enabled multi-stakeholder traceability reveals that integrating it with off-chain data validation, privacy preservation, and performance optimization allows the trust mechanism to facilitate supply chain transactions during data sharing [95]. Therefore, trustworthiness is not just an addition, but a basic structural constraint that needs to be prioritized in AEI design.

Essentially, the AEI technical framework is not just a pile-up of concepts, but the construction of a complete technical chain from physical entity coupling to intelligent collaborative optimization. It seamlessly integrates interconnected access, edge-layer autonomous control [91], platform layer data governance, digital twin strategy simulation, and artificial intelligence-driven multi-objective collaboration under uncertain conditions [93], connects virtual power plants (VPPs) with other market mechanisms, and the interaction methods of the mature microgrid market provide a reusable basis for large-scale commercial operations [102]. Eventually, this multi-layer framework transforms passive monitoring into active unified optimization, balancing crop yields, energy consumption, and low-carbon power generation [96]. This comprehensive system architecture relies on the integration of deep two-way information and energy flows, thereby providing a basis for the thorough investigation into the interactive coupling between agricultural informatization and electrification in later sections.

4. Coupling Analysis: Agricultural Informatization × Electrification

4.1. CPS Perspective on the Coupling of Agricultural Informatization and Electrification

The actual development situation of AEI. In agricultural scenarios like greenhouses, agriculture-photovoltaic complementary, and micro-grids, it is necessary to understand the

small mechanisms of how information and electricity are coupled, and also need to consider the super-macro technical architecture. From the perspective of CPS theory, agricultural intelligence is not only related to sensors or cloud dashboards; it also requires a powerful closed-loop system including observation, prediction, optimization, and execution. In this cycle, electrification has an impact on fine-tuning execution, and informatization has an impact on low-cost collaborative control. Without electrification, variables in agricultural production will be discrete or manual, and complex control inputs will be ineffective. Without informatization, electrified equipment can only rely on experience for local control and cannot rely on prediction, price, or multi-party collaboration for global optimization. This key interdependency is systematically addressed in the CPBS framework of protected agriculture and plant factories. This framework highlights the physical processes related to agricultural goals, such as heat and moisture, water, power systems, and the biological processes of crop physiology. The integrated models and controls at the biological, physical, and information levels need to be studied as a unified system to ensure crop yield and quality and reduce energy consumption and operating costs [103]. Thus, digital infrastructure and computer intelligence have prompted the transformation of funds from experience-driven agriculture to a dual-drive system engineering paradigm driven by data models [104].

4.2. Electrification Redefines Agriculture as a Schedulable Energy System

Electrification reshapes the CPS physical layer. Irrigation pumps, etc., are core actuators in agriculture. Cultivation is associated with quantifying power curves and constrained control variables. Taking irrigation as an example, the operation cost of pumping stations is closely related to the electricity price mechanism. So many studies obviously model irrigation scheduling as an optimization event under energy price constraints, such as near-real-time voltage reduction energy consumption optimization research [101], scheduling methods integrating real-time electricity prices and crop stress thresholds [105], and irrigation optimization methods using swarm intelligence algorithms to save a lot of money during high-price periods [106]. These studies show that electrification transforms irrigation from a simple water resources matter into a complex water-electricity coupled optimization matter. From a macro perspective, irrigation planning is required to balance the multi-criteria trade-off between agricultural output and water productivity, alongside holistic optimization and the advancement of integrated regional resource scheduling across water, electricity, and production systems [107]. And electrification not only provides drive for agricultural production but also explicitly incorporates agricultural processes into the constraints and objectives of the scheduling energy system.

4.3. Informatization Enables Predictable and Optimizable Flexible Loads

The informatization layer corresponds to the cyber layer of cyber-physical systems, with its core value lying in transforming the controllability of electrical equipment into system observability and computability. By integrating multi-source sensing data from the environment, soil, and crop conditions with power, energy, and equipment status information, this system generates sufficient state information required to support predictive and optimized decision-making. In precision irrigation, a robust perception-decision-execution closed-loop system has been established. Research suggests that integrating meteorological, soil, and crop-related monitoring plays a key role in improving water-use efficiency and strengthening automated scheduling stability [108], facilitating the technical evolution from simple threshold control to advanced model predictive control and optimal control [109]. Practical implementations, such as IoT-integrated machine-learning irrigation management systems [110] and intelligent irrigation systems utilizing LoRa-based data fusion and adap-

tive scheduling [111], systematically demonstrate how informatization converts irrigation power loads into predictable, optimizable, flexible resources [112]. This development trend aligns perfectly with the core concept of edge computing, empowering smart agriculture. Edge computing reduces transmission latency, decreases reliance on cloud bandwidth, and enhances system security, ensuring the stable operation of agricultural CPS even in network-constrained rural environments [91]. Recent reviews emphasize an edge-first framework—achieving device-level autonomous control by combining lightweight deep learning models with rule-driven decision logic [76], which, alongside edge-cloud collaborative IoT architectures [92], ensures that the flexible loads enabled by electrification can be reliably and optimally dispatched.

4.4. Greenhouse CPBS for Unified Crop-Energy Optimization

Greenhouses and factories are one of the most electrified agricultural scenes, and are the main examples of the centralized embodiment of CPS (controlled physical system) and CPB (controlled physical business system). Its key energy and heat demand across heating, cooling, ventilation, dehumidification, auxiliary lighting, and CO₂ crop growth are strictly controlled by microclimate conditions. So, energy conservation should not damage the output or quality. To address this issue, numerous existing studies have formulated greenhouse control as a constrained optimization problem, with predictive control and its enhanced algorithms serving as the core methodology. For example, the research on MPC (model predictive control) for Venlo greenhouse clearly aims to cut down on energy, water, and carbon dioxide consumption while maintaining the best crop growth conditions [113]. Hierarchical MPC research further emphasizes improving energy efficiency through stratified strategies to promote clean production [114]. Data-driven robust MPC frameworks balance control performance and energy consumption evaluation while addressing meteorological uncertainties and model errors [115]. These findings collectively demonstrate how modeling, prediction, and optimization enable electrified controllable devices like HVAC and lighting systems to evolve from localized control to system-wide optimal control.

In engineering practice, the observability of greenhouse systems heavily relies on wireless sensor networks and spatially distributed monitoring technologies. A comprehensive end-to-end review of greenhouse wireless sensor networks systematically examines network deployment strategies, sampling approaches, and decision support models [116]. Research on greenhouse climate and crop condition monitoring further indicates that environmental spatial heterogeneity increases the difficulty of optimal control, necessitating multi-point monitoring to obtain more precise state characterization [117]. Zigbee-based greenhouse data acquisition systems and wireless data fusion studies for temperature monitoring also validate the “sensor quality determines control quality” chain reaction [118].

4.5. Agrivoltaics for Land-Energy-Crop Optimization

Agro-PV technology has propelled the supply side of agricultural electrification into a new phase of “local power generation and local consumption”: the agricultural system is no longer merely a power load but also a symbiotic carrier of renewable energy assets. Long-term basic research has demonstrated that agro-PV technology can enhance land-use efficiency to achieve energy production while alleviating the competition between energy and arable land [119]. Recent reviews have systematically organized the knowledge system of agro-PV technology from perspectives such as agronomic mechanisms, system design parameters, and future development directions [120], positioning it as a strategic tool to balance food and energy supply. Specifically, studies combining agro-PV engineering design with smart agriculture indicate that it is essential to rely on information-based mon-

itoring, modeling, and control to manage the relationships between shading, microclimate, and crop physiological responses, thereby achieving synergistic optimization of power generation and crop yield/quality [121]. Quantitative research on China's PV development and arable land protection further suggests that agro-PV technology may mitigate land-use conflicts, but requires refined evaluation and planning [122]. Therefore, agro-PV technology clearly demonstrates the technical path: "energy system and market, carbon signals → information layer → load and equipment scheduling". Electrification embeds energy supply into agricultural production space, while informatization transforms the multi-objective trade-offs between land, energy, and crops into an optimizable closed-loop decision.

4.6. Microgrids and VPPs for Agricultural Energy Coordination

When agricultural electrification loads (such as greenhouses, irrigation, and cold chain systems) are integrated with distributed energy sources like PV, energy storage, and biomass to form campus or rural microgrids, the system research paradigm shifts from equipment-level control to operational scheduling and collaborative optimization. A seminal review on MEMS provides methodological foundations for energy-side research in agricultural contexts through comparative analysis of decision-making strategies, solutions, and uncertainty modeling [98]. Subsequent reviews systematically organize MEMS research findings across dimensions, including strategic planning, predictive analytics, control technologies, pricing mechanisms, and economic principles [123] while conducting comparative studies on renewable energy intermittency and demand response [124]. The relevance between this research direction and agriculture lies in the fact that electrification transforms agricultural energy flow into adjustable variables, while informatization enables agricultural systems to participate in a broader system coordination.

In addition, virtual VPP is the core interface mechanism to integrate decentralized agricultural energy assets and controllable loads into market trading entities. This paper systematically summarizes the market-oriented VPP model, and classifies the modeling method, solution, participation mode, and implementation feasibility [102]. Recent studies on VPPs and virtual aggregators have expanded into auxiliary services, control methodologies, and revenue mechanisms [125], whilst simultaneously analyzing the layout of virtual power plants (VPPs) and their interactive relationships with power markets, carbon trading markets, and flexibility provision markets [126]. These studies jointly support a system-level conclusion that forecasting, aggregation, optimization, and settlement capabilities are crucial to upgrading agricultural electrification load from "power users" to "tradable flexible resources".

Another issue that must be clarified is the parallel operation of distributed generation and centralized power grids. In actual AEI deployment, photovoltaic, wind, biomass power generation, energy storage, and flexible agricultural loads typically operate in grid-connected mode rather than completely isolated. This requires coordinated handling of anti-islanding protection, synchronization, fault ride-through, voltage support, harmonic control, and scheduling coordination between local controllers and utility side requirements. Therefore, reliability, power quality, and electrical safety are not secondary engineering details, but core design standards.

4.7. Information-Driven Energy Management for Electrified Agricultural Machinery

Another key direction of agricultural electrification is the electrification of agricultural machinery and mobile equipment. The working range, efficiency, and power-response capability of electric tractors and similar equipment are strongly determined by energy-management strategies, forming a typical CPS closed-loop system of driver-state recognition, energy allocation, control execution, and feedback learning. In studies

on reinforcement-learning-based energy management of electric tractors, power demand is modeled as a Markov process, and the adaptability of energy allocation is enhanced through driver-status identification [127]. For a hybrid energy storage system, a two-layer adaptive power management strategy and a multi-objective energy management optimization framework integrate battery life, operation efficiency, and stability into a unified optimization objective [128]. Additionally, the application of reinforcement learning in electric tractor drive control demonstrates potential to improve traction efficiency and reduce energy consumption [129]. These studies collectively indicate that electrification provides controllable power systems for farm machinery, while information capabilities such as recognition, learning, and optimization determine whether equipment can achieve efficient and reliable operation under complex field conditions.

4.8. Unified CPS View of Informatization-Electrification Coupling

In summary, the intrinsic connection between agricultural informatization and electrification can be comprehensively explained through the CPS-coupled closed-loop system, thereby enhancing system controllability and flexibility. Informatization, through sensing technologies, edge cloud computing, modeling, digital twins, and optimal control, converts these controllable variables into predictable, verifiable, and collaboratively optimized decision-making schemes, improving system observability, collaborative capabilities, and optimality. In irrigation, the coupling of electricity pricing/market signals with pump scheduling has made “water-electricity” joint optimization the mainstream development direction [130]. In greenhouse and protected agriculture, the CPBS framework integrates crop physiological characteristics with energy control modeling, forming an interpretable energy-saving closed-loop through MPC and intelligent RMPC [131]. In agrivoltaics, microgrids, and VPP scenarios, electrification expands the spatial scope of energy supply and flexible resources, while informatization provides capabilities for cross-entity collaborative scheduling and market interaction [132]. Regarding agricultural machinery electrification, learning-based energy management enables adaptive optimization capabilities for electrified equipment under complex operating conditions [133]. Therefore, the situation of two-way interdependence is consistent. The situation of informatization and electrification is not something that occurs one after another, but promotes each other: a higher degree of electrification helps to optimize informatization, and stronger informatization makes electrification assets have flexible characteristics.

The coupling of information and electricity theory lays the foundation for AEI. Nevertheless, three major technical constraints need to be resolved before this concept can be implemented on a broad scale. First, deep agricultural decarbonization requires extending traditional electricity to the production of high-value green fuels, and advanced energy conversion technologies such as SOEC need to be used to deal with the problem of seasonal energy surplus. Second, to achieve ultra-low-latency on-site execution without relying on vulnerable cloud networks, it is necessary to shift from passive perception to closed-loop operation of edge AI-IoT. Third, when AEI is extended to multi-stakeholder systems, reducing data leakage and ensuring reliable cross-domain optimization, strong privacy, security, and trust mechanisms in federated edge intelligence are urgently needed. These three frontier directions collectively constitute the critical technological breakthroughs.

4.9. Explanatory Regional Cases: Seasonal Balance, Reliability, and Multi-Departmental Coupling

The AEI case can be represented as a county-level agricultural park covering protected horticulture, field irrigation, livestock breeding, cold-chain warehousing, and primary agricultural product processing.

From the energy supply side, feasible investment portfolios are not limited to photovoltaic power generation. According to local conditions, the system can integrate distributed photovoltaics, wind power, small-scale hydropower, biomass/biogas derived from livestock manure and crop residues, as well as electrochemical energy storage and thermal storage devices, while operating interactively with the distribution network. The core principle is that the energy architecture design must adapt to local climate, terrain, water resource endowments, agricultural and industrial structure, and policy constraints.

Regional energy demand exhibits obvious seasonal differences. In summer, the load is dominated by irrigation pumping, cooling, ventilation, and cold-chain consumption. In winter, greenhouse heating, constant-temperature regulation for livestock, and warehousing energy demand become more prominent. During the harvesting and processing season, temporary load peaks emerge in drying, sorting, refrigeration, and logistics links. Therefore, AEI analysis must simultaneously consider electric load curves, thermal load curves, and process energy intensity, rather than relying merely on annual total energy consumption for generalized evaluation.

The same park case also reflects the coordinated operation of distributed and centralized power systems. Under normal grid-connected mode, local photovoltaic, wind power, biomass/biogas combined heat and power, and energy storage systems jointly supply power to various agricultural loads together with the main distribution network. When renewable energy generation is surplus, the park can export electricity or participate in demand response through aggregators. In the event of grid disturbances or peak electricity price periods, critical loads such as ventilation, cold-chain supply, livestock water supply, and greenhouse environmental regulation should be guaranteed an uninterrupted power supply via energy storage or islanded microgrid operation. This grid-connected-islanded coordinated operation mode indicates that AEI requires not only the scheduling capability of distributed generation but also interface adaptation with the centralized power grid system.

The degree of reliability dependence varies significantly across different scenarios. A short-term power outage only leads to pump shutdown and minor losses, while a prolonged blackout may cause severe imbalance of greenhouse microclimate, pressure on livestock living conditions, milk deterioration, incubation equipment shutdown, and spoilage of post-harvest agricultural products. Accordingly, outage loss cost, allowable interruption duration, and agricultural product safety risks should all be set as explicit optimization constraints. This fundamentally explains why information-energy closed-loop control, fast local edge response, and coordinated parallel operation with the power grid have become cutting-edge development requirements for agricultural energy systems. This information-electrification coupling relationship is schematically shown in Figure 4.

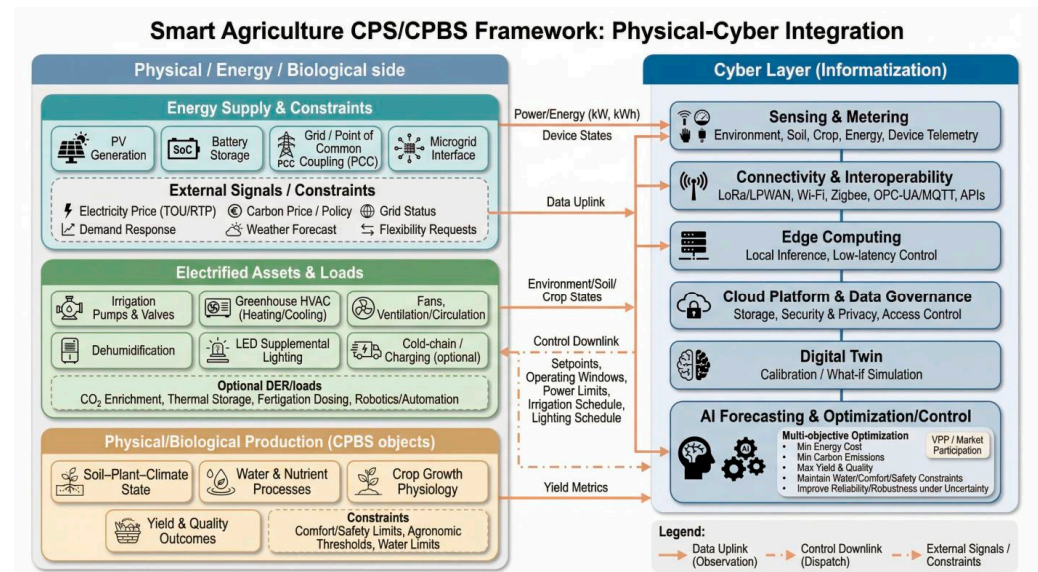


Figure 4. Schematic diagram of the relationship between agricultural informatization and agricultural electrification.

5. Three Frontier Directions and Their Frontier Significance

The cutting-edge directions discussed in this section are selected based on the following four criteria: (1) they must address structural bottlenecks that traditional single domain optimization cannot solve (2) they must create new coupling opportunities between agriculture, energy, and intelligent control (3) they must have the potential to reshape system architecture, not just improve individual components (4) they must be relevant to future scalable, low-carbon, and resilient deployments. This section aims to list the cutting-edge technologies in this field and explain the reasons why they have become the forefront representatives in the AEI field.

5.1. High-Temperature Solid Oxide Electrolysis at 750 °C

The forefront of this technology lies in its introduction of a new pathway for converting surplus renewable electricity and heat resources into storable, transportable, and saleable energy carriers. In highly electrified agricultural parks, this can transform abandoned wind management, seasonal storage, and carbon value realization from passive constraints to active system options. The integration pathway of 750 °C SOEC within AEI is illustrated in Figure 5.

In the advanced development of AEI, electrification is not only applied to irrigation, greenhouse regulation, cold chain, and agricultural equipment. A key trend is to convert the excess renewable electrical energy into transportation energy, storage, and production of green fuels, so that the value chain of agricultural electricity is expanded from simple consumption to the conversion of electricity to fuel. At this time, SOECs operating at about 750 °C are a promising method for efficiently producing green hydrogen. Different from traditional low-temperature water electrolysis, the SOEC technology uses heat to provide part of the reaction energy, reduces power consumption, and improves efficiency. This makes SOECs suitable for agricultural energy systems combined with PV, heat demand, and flexible loads. Recent research [134] shows that electrode engineering, interface optimization, and oxygen electrode materials are all changing, but the performance of SOEC still has problems. Challenges such as long-term durability, redox stability, and thermal stress management are still key important issues. Within the AEI framework, SOEC is not only used for hydrogen production but can also connect renewable energy, waste heat, and hydrogen demand together to form an optimized energy conversion unit.

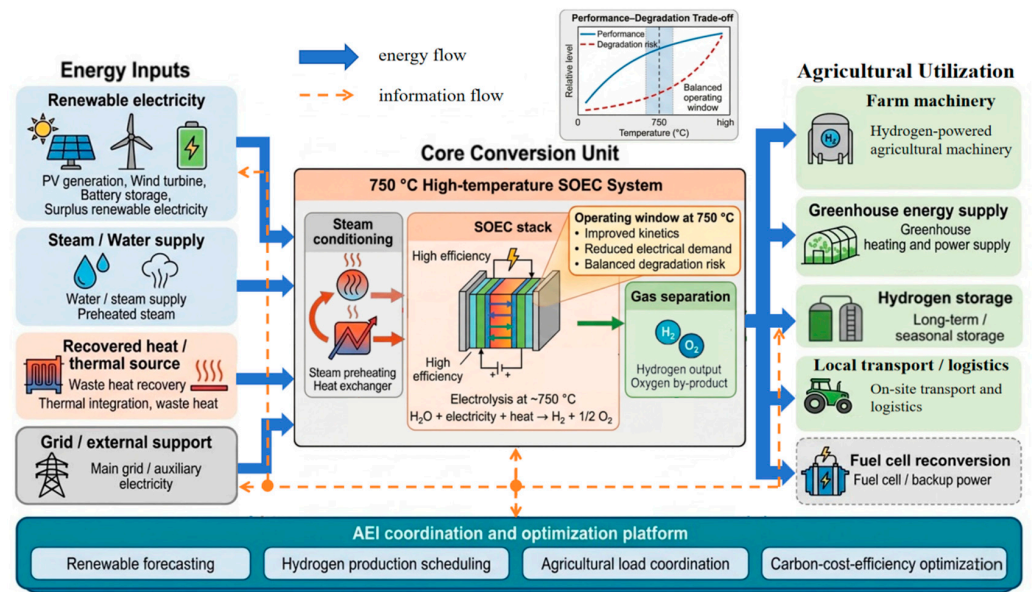


Figure 5. Schematic diagram of 750 °C SOEC integration in the AEI.

Sassone et al.'s work further emphasizes the importance of working temperature. They studied the effect of operating temperature on SOEC degradation by comparing the long-term performance at 750 °C, 800 °C, and 850 °C [135]. Their research findings indicate that temperature is not only a performance improvement parameter, but also a key variable that determines the trade-off between efficiency and durability. On the one hand, increasing the temperature typically improves electrochemical kinetics, reduces polarization losses, and supports higher hydrogen production efficiency. On the flip side, extremely high working temperatures tend to speed up the degradation procedure, including nickel depletion, interface diffusion, and unfavorable microstructural evolution of electrode materials [135]. In this case, 750 °C represents an important operating window. It is high enough to maintain the thermodynamic and kinetic advantages of steam electrolysis. But compared to more aggressive high-temperature conditions, it is gentle enough to reduce material degradation. As a result, from an engineering perspective, 750 °C SOEC can be understood as a compromise between performance, efficiency, and long-term stability. This is particularly valuable for distributed agricultural applications that require robust and continuous operation.

In the AEI scenario, the integrated value of SOECs at 750 °C has three main aspects. The first is to find new ways to consume local excess renewable electricity. In the farm/park agricultural system, the PV output is relatively intermittent, and the excess electricity at noon does not match the agricultural load. Introducing SOECs converts excess electrons into hydrogen (H₂), breaking through the limitations of direct power storage and expanding the flexibility of the use time of renewable energy. The second is that SOECs have inherent thermal integration compatibility. Agricultural facilities, such as protective agriculture and processing systems, have similar heat demand and waste heat release situations. SOECs require high temperatures, which can enable the reutilization of available heat, improve the efficiency of the system, and build a coordinated framework of “renewable power + heat recovery + H₂ production”. Third, producing hydrogen is a commonly used energy carrier in agriculture, which can be used for backup power of fuel cells, energy supply for combined heat and power in greenhouses, agricultural transportation fuels, and seasonal energy storage. The key is that the development of SOECs makes there be more cooperative options for the AEI, expanding from traditional power load scheduling to comprehensive electricity-heat-H₂-chemicals management.

Agricultural 750 °C SOECs have great potential but also significant engineering challenges. These challenges include maintaining durability during fluctuations or medium-term periods of rural renewable energy (RE) systems, handling complex issues of system integration (such as steam, startup, hydrogen treatment, etc.), improving economic feasibility, and conducting capital cost comparisons. However, 750 °C high-temperature SOECs are at the forefront of the transformation for agricultural energy systems to shift from direct electrification to the production of high-value green fuels. Initiatives to strengthen the integration of electricity-heat-fuel have broadened the functional scope of AEI and opened up the possibility of new low-carbon, automated, and flexible farm energy systems, thus becoming a potentially transformative infrastructure for the next generation of smart agriculture [134,135].

5.2. Edge AI-IoT Agricultural Closed-Loop Operations

Agricultural intelligence cannot create true operational value unless perception, prediction, decision-making, and execution are completed in a reliable local control loop. Edge artificial intelligence IoT closed-loop operation, as a cutting-edge direction, can shift AEI from visual monitoring and pure cloud analysis to low latency, interrupt-resistant, and on-site deployable autonomy.

Compared with open-loop monitoring, closed-loop AEI technology requires a complete chain of sensing, prediction, decision, actuation, and feedback. The controller should continuously compare the predicted state with the actual microclimate, soil moisture, livestock environment, or energy status, and then update irrigation, ventilation, lighting, heating/cooling, storage charging/discharging, and demand-response commands. In this sense, closed-loop technology is the mechanism that converts digital observation into real production and energy-saving effects. The corresponding AI-IoT closed-loop operation process is summarized in Figure 6.

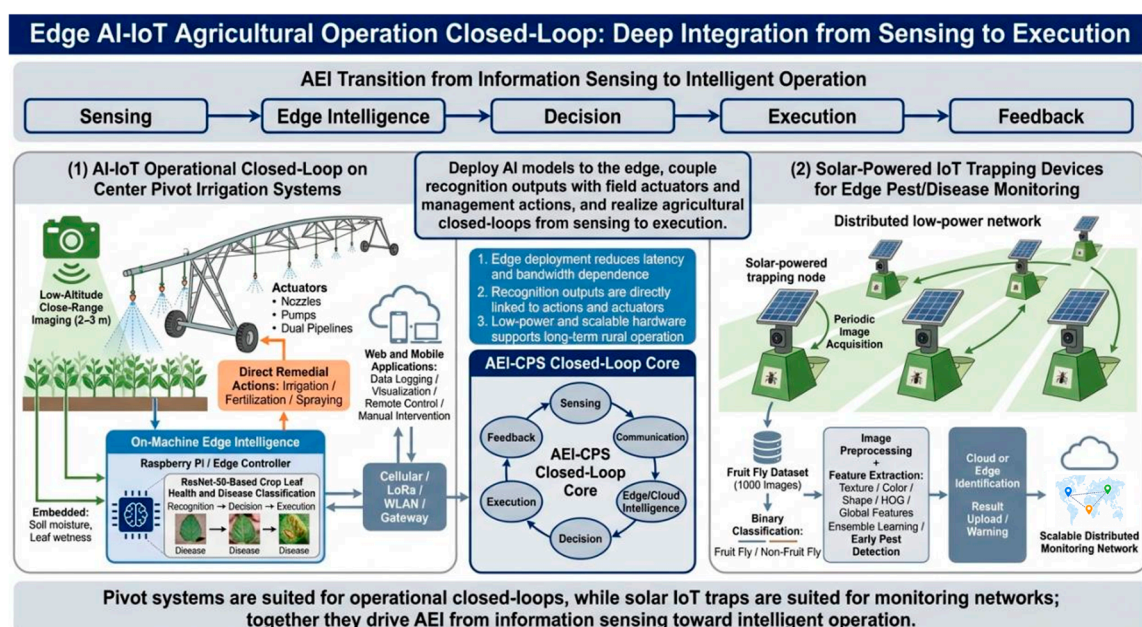


Figure 6. AI-IoT Agricultural Operations Closed Loop.

In the cutting-edge applications of AEI, the core focus of intelligent equipment lies in integrating information perception with operational execution into a closed-loop system, enabling the transition from “identification and early warning” to “decision-driven operations”. This is specifically manifested through two typical carriers:

(1) Edge-Based Closed-Loop Irrigation System.

This study constructed an “Agricultural AI IoT Intelligent Hub System” by deploying image processing and deep learning models on the central hub irrigation system, achieving crop image acquisition and 2–3 m low-altitude spraying and irrigation operations. Thus, bypassing the restrictions of drones [136]. The system uses the ResNet-50 model for leaf health and disease type classification. Directly control the pivot actuator based on the recognition results to achieve a closed-loop “recognition decision execution”. In terms of architecture, the communication layer uploads data to the cloud or application layer through cellular or LoRa/WLAN gateways. The application tier provides access ports supporting data archiving, graphical visualization, remote control functions, and human-assisted intervention. The hardware layer includes a controller, actuator, multipoint camera, and a communication module. The edge computing function is realized through Raspberry Pi and real-time model deployment on-site. This system deeply integrates agricultural electrification (powertrain and actuator) with informatization (identification model and communication technology), establishing a sustainable closed-loop CPS.

(2) Edge-Based Low-Power Pest Monitoring System

This study utilizes solar-powered IoT insect traps and captures images through cameras and sensors for initial edge-level recognition. Finally, upload the results to the cloud for administrator notification [137]. A fruit-fly image dataset containing 1000 images for binary classification (fruit fly/non-fruit fly) was constructed, covering different angles, lighting conditions, and time periods to improve model robustness. The processing follows an edge-computing workflow of feature extraction and classification, including image preprocessing, texture, color, and shape feature extraction, and ensemble-learning classifier comparison. The results indicate that integrating multiple features and models significantly improves recognition accuracy.

In conclusion, the cutting-edge research in this paper goes beyond the simple application of AI in agriculture, focusing instead on the three core requirements of the closed-loop AEI system: (a) deploying intelligent models to edge devices to reduce transmission latency and bandwidth dependency, (b) directly coupling recognition output with field actuators to achieve closed-loop operation control, (c) using low-power, expandable hardware to enable long-term field operation and wide area coverage, and (d) leveraging the pivot system for closed-loop control scenarios while the IoT insect-trapping device excels in monitoring networks. Together, these elements form the core pathway for agricultural intelligence to transition from information perception to intelligent operation.

5.3. Privacy, Security, and Trust in Joint Edge Intelligence

Due to AEI’s increasing reliance on cross-stakeholder data exchange, federal intelligence, and platform-based coordination. Once agricultural equipment, energy assets, and market entities are connected, the value of the system can only expand when collaboration remains auditable, privacy protected, network resilient, and trustworthy in terms of institutions. Therefore, privacy, security, and trust have become one of the forefront directions in this field. The layered privacy-security-trust architecture for this federated edge-intelligence scenario is shown in Figure 7.

As the AEI advances into the “multi-stakeholder collaboration + edge intelligence closed-loop” phase, systemic risks have evolved beyond algorithmic inaccuracies to include data breaches, model poisoning, unauthorized access, and trust deficits in cross-domain collaboration. This challenge is particularly acute in agricultural contexts, where sensitive data—including crop yields, land/field operations, and equipment status—is frequently transmitted through vulnerable networks, heterogeneous devices, and multiple stakeholders (farmers, cooperatives, platforms, and equipment suppliers). This section, therefore,

focuses on integrated “privacy-security-trust” mechanisms, which are essential for enabling scalable deployment and practical implementation of both the third-phase technology stack and the fourth-phase CPS system.

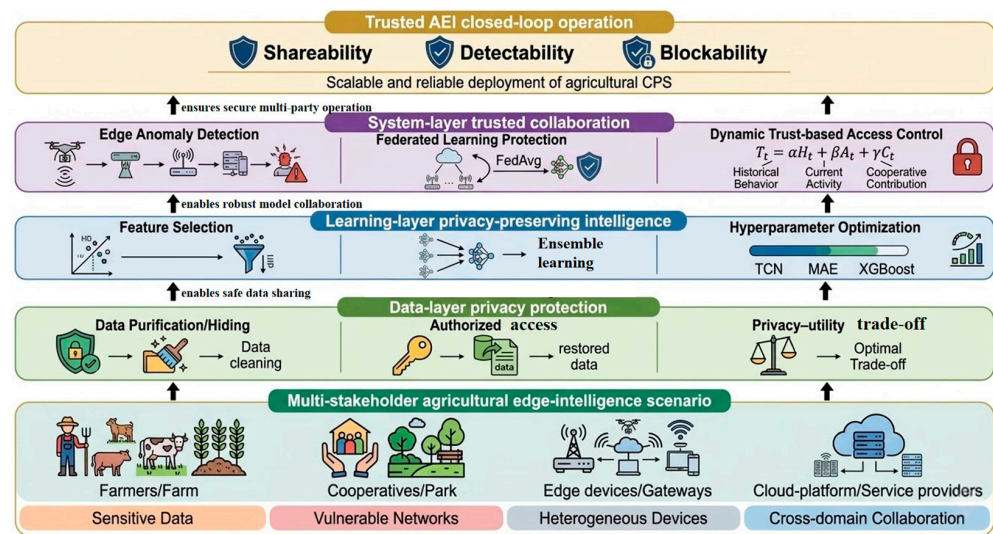


Figure 7. Layered privacy-security-trust architecture for federated edge intelligence in AEI.

(1) Data-Layer Privacy Protection

A two-stage privacy protection framework combining “data purification and data restoration” optimizes the generation of “optimal keys” to maintain a balance between the extent of sensitive data obfuscation and the practical usability of data. Traditional IIoT authentication and privacy policies often face adaptability issues due to factors such as single-factor authentication, expanding user base, and diverse user demographics. To address this issue, this study models privacy protection as a constrained multi-objective optimization problem. The evaluation indicators include “modification level, hiding rate, correlation between raw data and restored data, and information retention rate” [138]. This approach establishes a computable balance between privacy preservation and practical value.

The study also proposed using the Grasshopper Black Hole Optimization (G-BHO) algorithm to generate optimal keys, claiming that its privacy metrics have improved compared to multiple benchmark algorithms [138]. In terms of implementation, it further specifies that only authorized parties can use the same optimal key to perform “hide” and “recover” operations [138], thereby embedding access control logic directly into the privacy workflow itself.

(2) Learning-Layer Privacy Preservation

In high-dimensional IoT big data environments. Relevant findings indicate that the continuous expansion of IoT infrastructures renders high-dimensional data risks and privacy concerns more conspicuous than ever before. However, traditional server-assisted methods have shown limitations in solving collusion attacks and complex dynamic environments.

To address this issue, the study proposed PPSLOA HDBDE. Firstly, it applies a linear scaling normalization technique, then uses the Shamao group optimizer for feature selection to alleviate high-dimensional problems, followed by intrusion detection through ensemble models (TCN, MAE, and XGBoost), and finally uses an improved version of the ocean predator algorithm to optimize the hyperparameters of the three models, achieving an accuracy of 99.49% [139].

The AEI study demonstrates that agricultural marginal environments generate high-dimensional heterogeneous data (e.g., images, sensor streams, operational logs, energy consumption data, etc.). Single models often struggle to maintain robustness under privacy constraints. Therefore, compared to isolated single-model solutions, a systematic approach combining “optimization-driven feature selection, ensemble learning, and fine-tuning” proves more practical for real-world deployment as a reusable privacy-preserving learning module.

(3) System-Level Trusted Collaboration

When AI enterprises truly engage in cross-stakeholder collaboration, relying solely on data purification or privacy-preserving learning remains insufficient to address practical challenges such as model poisoning, unauthorized access, and low-latency requirements. Research clearly indicates that many IoT security solutions rely on cloud processing. This may limit detection capabilities and increase latency when migrating to edge computing. Meanwhile, although federated learning has achieved distributed training, it still faces challenges such as non-IID data, device heterogeneity, and gradient leakage. In addition, static access control cannot reflect the dynamic changes in device behavior and context [140]. To address these limitations, the AI-SET architecture proposes a unified solution. This scheme integrates three modules—resource-efficient edge anomaly recognition, differential private federated learning with homomorphic encryption, and adaptive access control based on trust metrics.

From a mechanical perspective, AI-SET not only defines module boundaries but also provides an operable standardized workflow. In the federated learning stage, it uses differential privacy techniques to perturb gradients, combines homomorphic encryption to ensure consistency of encryption updates, and uses a federated averaging algorithm for data aggregation. In the access control phase, the framework defines a trust score $T_t = \alpha H_t + \beta A_t + \gamma C_t$, integrating historical behavior, anomaly ratings, and contextual factors, and makes access decisions through dynamic thresholds [140]. In experimental implementation, the framework emphasizes a test platform incorporating edge devices such as Raspberry Pi and Jetson Nano, together with TensorFlow federated learning and toolchains such as PySyft and TenSEAL, demonstrating a deployable and reproducible engineering approach [140]. In terms of performance, AI-SET achieves 94.3% anomaly-detection accuracy and a 3.2% false-positive rate, while its trust-based access control reaches a 96.8% blocking rate for unauthorized access attempts [140].

More specifically, the comparative analysis reports blocking rates at different thresholds [139], accuracy variations across training rounds, and systematic differences in bandwidth, latency, and interception rates compared to cloud-based centralized federated learning (FL) and edge intrusion detection system (IDS) baselines—providing quantitative evidence for “trusted edge federated intelligence”.

The core breakthrough in this section lies in upgrading the AEI from a “runnable system” to one that “operates stably and reliably in real-world multi-party environments over extended periods”. Specifically, privacy protection (data/learning) addresses “shareability”, security detection tackles “detectability”, and trust control resolves “blockability”. These mechanisms work in concert to ensure the agricultural CPS closed-loop system remains immune to attacks or misuse in open environments, thereby enabling the large-scale deployment of the third-party technology stack.

6. How AEI Enables Smart Agriculture

Having explored the overarching architecture, specific coupling scenarios, and frontier technologies, there is a need to conduct a macroscopic analysis of the manner in which this AEI paradigm structurally enhances the fundamental characteristics of smart agriculture.

AI has propelled the development of smart agriculture into a new stage. At present, the era of artificial intelligence is promoting the evolution of the system to cognition, prediction, autonomous optimization, and continuous learning. In essence, artificial intelligence provides an “intelligent brain” for Smart Agriculture, while the AEI provides “computable physical carrier” and “controllable energy infrastructure”, enabling intelligent decision-making to be transformed into reliable action. On the one hand, artificial intelligence enhances the perception, understanding, and prediction ability of agricultural systems in complex environments by integrating multimodal data such as weather, soil, crop phenotype, power load, PV output, and equipment status. On the other hand, the AEI transforms agricultural processes such as irrigation, greenhouse environmental control, light supplement, cold chain operation, and even hydrogen production and hydrogen storage into related schedulable energy processes.

Therefore, the combination of artificial intelligence and AEI is not a simple superposition of technologies, but a deep coupling of intelligence and electrification. In contrast, under the framework of AEI, artificial intelligence acts on an integrated cyber-physical-biological system, in which agricultural load, distributed energy, energy storage, environmental control equipment, and digital twins are all incorporated into a unified decision-making closed loop. In this way, AI can not only support single-task prediction but also realize cross-domain collaborative optimization.

At the same time, the era of artificial intelligence also puts forward new technical requirements for AEI. As artificial intelligence is deeply embedded in agricultural energy systems, data quality and data governance become more critical. Stable AI operation is strongly determined by unified data standards, traceable records, and sustained long-term credibility of datasets. However, agricultural and energy data are naturally heterogeneous, multi-scale, and noisy, and are often provided by decentralized entities with different interest demands and reliability. This means that the future AEI platform must not only collect data but also provide standardized governance, calibration, version control, uncertainty labeling, and auditability. In other words, data governance in the AEI is not only an engineering support function, but also a prerequisite for the introduction of reliable artificial intelligence. Without high-quality data governance, AI models may not reduce uncertainty but amplify noise, drift, and bias.

The introduction of AI into the AEI is not to pursue unrestricted and complete autonomy, but to build a system with adaptive and interpretable decision-making capabilities within the boundaries of agriculture, energy, economy, and ecology. This is why digital twins, a mechanism for data fusion model and an interpretable optimization framework, are particularly important. Digital twin can provide simulation verification before field deployment; the mechanism-based AI can ensure the physical consistency between the agricultural side and the energy side; Interpretable control helps users understand the reasons behind the recommended scheduling behavior.

Not only by replacing energy, but AEI can also reduce carbon footprint by improving the agricultural production structure. This includes transferring high-energy operations to periods of surplus renewable energy, selecting crop or livestock production modes with better energy output ratios, coordinating greenhouse planting plans based on seasonal heat and light availability, integrating crop and livestock waste energy cycling systems, and using digital twins to compare alternative production structures before implementation. These measures reduce carbon intensity through energy substitution and production structure optimization.

Artificial intelligence technologies have reshaped the value framework of the AEI. Meanwhile, they have facilitated the transition of smart agriculture—shifting from independent farm management modules to comprehensive operational ecosystems linked to

the full-scale energy market and carbon trading market [141]. With the prediction and optimization capabilities enhanced by artificial intelligence, the AEI can aggregate flexible agricultural loads and decentralized energy into operational assets that can be dispatched externally. This enables farms, greenhouses, agricultural parks, and rural cooperatives to transform from passive power users to active participants in demand response, flexible services, virtual power plants, and the future carbon perception market. Artificial intelligence plays a central role in dealing with uncertainty, rapid scheduling, and multi-objective decision-making, while the AEI provides the asset structure and communication control chain necessary for participating in the market. It can be seen that AI + AEI is likely to redefine the economic role of agriculture in the energy transition, so that the agricultural system is not only a food producer, but also a flexible low-carbon infrastructure.

Nevertheless, the large-scale implementation of AI + AEI still faces a number of practical challenges. First, cross-domain modeling is still difficult. Because agricultural load is not a rigid load, and its control effect depends on crop growth stage, environmental threshold, and management objectives, the pure data-driven model has limitations and must integrate biological and physical knowledge. Secondly, AI models must operate in harsh real environments. In addition, as the AEI evolves into an intelligent infrastructure with multi-agent participation, privacy protection, network security, access control, and traceability governance will no longer be optional additional functions, but will become basic requirements. The economic feasibility of the application cannot be ignored. The AI + AEI system should not only reflect the advanced technology, but also reflect its full life cycle value under the constraints of real farm profitability, energy price fluctuation, and operation and maintenance ability.

On this basis, compared with fully centralized control, the layered deployment architecture of “edge autonomy + cloud intelligence + platform governance + Digital twin verification” is more realistic and feasible. This architecture can not only ensure the stability of the local closed-loop but also benefit from cloud model training, cross-scene knowledge migration, and global optimization.

As summarized above, amid the contemporary era of artificial intelligence, the value of the AEI is to narrow the gap between theoretical intelligence and on-the-ground implementation in smart agricultural systems. Artificial intelligence gives agricultural systems a stronger ability to perceive, predict, and adapt, while AEI makes these systems more controllable, schedulable, and integrated. The combination of the two is likely to become the most critical path to promote Smart Agriculture from digital monitoring to intelligent, low-carbon, and resilient system operation.

7. Conclusions

In summary, AEI is not simply a continuation of agricultural informatization or rural electrification, but a new system paradigm for the deep integration of agriculture, environment, energy, and intelligent decision-making in the AI era. Based on the literature reviewed in this paper, the following conclusions can be drawn:

(1) AEI: Enabling Intelligent Closed-Loop Agriculture

Through the integration of controllable agricultural loads, distributed renewable energy, energy storage, sensing networks, edge computing, and digital platforms, AEI transforms agricultural production processes into computable and schedulable cyber-physical systems. This enables smart agriculture to move beyond visual monitoring and fragmented automation toward coordinated optimization of production, energy, cost, carbon, and reliability.

(2) Agri Info-Energy Integration: Foundation of AEI

Electrification converts irrigation, greenhouse environmental control, lighting, machinery, and other agricultural processes into adjustable energy-linked actuation chains, while informatization provides observability, prediction, communication, and optimization capabilities. Their interaction forms the technical basis for agricultural cyber-physical-biological systems and explains why the value of informatization rises with increasing electrification, while the flexible value of electrification is released through stronger information capabilities.

(3) AI + AEI: Pathway to Next-Gen Smart Agriculture

AI strengthens AEI by enabling multimodal perception, uncertainty prediction, anomaly diagnosis, adaptive control, and multi-objective optimization, while AEI provides AI with real execution scenarios, controllable assets, and cross-domain coupling boundaries. As a result, smart agriculture is evolving from task-level intelligence toward system-level intelligence in which AI can directly support operational decisions under agronomic, energy, and carbon constraints.

(4) Key Enabling Technologies for Scalable AEI

Because AEI involves heterogeneous, noisy, and multi-stakeholder data flows, future development depends on standardized data governance, quality control, traceability, and auditable sharing mechanisms. At the same time, digital twins and edge-cloud collaborative architectures are essential for achieving simulation-based validation, local autonomous control, and cloud-level model training and optimization, thereby supporting both safety and practical operability.

(5) Challenges for Scaling AEI

The coexistence of weather uncertainty, renewable intermittency, agricultural seasonal load variability, and fragile rural infrastructure makes AEI operation significantly more complex than conventional agricultural automation systems. In addition, storage costs, maintenance capabilities, privacy protection, network security, access control, and trust mechanisms will directly determine whether AEI can transition from technically feasible pilot projects to economically sustainable and socially acceptable real-world systems.

(6) Future Directions for AEI Deployment

Highly electrified and controllable scenarios such as greenhouses, precision irrigation, protective agriculture, and microgrids in agricultural parks may be the most promising areas for early deployment. Starting from these scenarios, gradually integrating market mechanisms such as artificial intelligence, digital twins, carbon accounting, demand response, and virtual power plants may provide a realistic path for standardization, scalability, and widespread adoption.

To sum up, the AEI should be regarded as the strategic technical framework for future smart agriculture rather than an isolated engineering solution. Its true significance lies in making agricultural development a more intelligent, low-carbon, resilient, and collaborative optimization system. In this system, food production, energy management, and environmental governance are no longer treated separately, but are coordinated within a unified decision-making framework.

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